

Hidden causality in Modern Greek

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Abstract

This paper explores the syntax and semantics of an attitudinal construction in Modern Greek, where an attitude verb takes an accusative object followed by a complement clause. Building on existing syntactic literature (e.g., [Hadjivassiliou et al. \(2000\)](#); [Kotzoglou \(2002\)](#); [Kotzoglou and Papangeli \(2007\)](#); [Kotzoglou \(2017\)](#)), I show that the accusative object is base-generated higher than the lower clause. Yet, I show that it semantically behaves as if it is part of the intensionalized argument of the attitude verb, giving rise to *de dicto* readings ([Tsilia, 2023](#)). Building on this and on a causal semantic requirement associated with the accusative object, I suggest a clausal analysis of the phenomenon. More specifically, under this analysis the accusative object is the subject of a small intermediate VP clause headed by a silent proleptic CAUSE, which then takes the complement clause as its object. This contributes to the literature suggesting that hidden clauses are cross-linguistically attested and can solve intensionality paradoxes ([den Dikken et al., 2018](#)), as well as to the literature on prolepsis ([Davies, 2005](#); [Salzmann, 2017a](#); [Deal, 2018](#); [Dawson and Deal, 2019](#)) showing that proleptic constructions may have varying interpretations and syntactic analyses cross-linguistically.

Keywords: syntax-semantics interface, prolepsis, intensionality, concealed causatives, causal constructions, quasi-ECM, hidden clauses, Modern Greek

1 Introduction

Modern Greek (MG) features attitudinal constructions where verbs like *thelo* ‘want’ take an accusative object (ACC DP) followed by a CP containing a pronoun or a gap. [Hadjivassiliou et al. \(2000\)](#); [Kotzoglou \(2002\)](#) dub this the “quasi-ECM” construction. Here is the basic

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pattern:¹

- (1) I Maria theli **ton** **Yani** [na aghapai *pro* mono
The.NOM Maria.NOM want.3SG.PRS the.ACC Yanis.ACC SBJV love.3SG 3SG only
 aftin].
 her.ACC
‘Maria wants Yanis to only love her.’

Tsilia, 2023: p.656, ex.6

Notably, “quasi-ECM” has an alternative where the DP remains in the CP and bears NOM case. The NOM DP appears in SpecCP in a topicalized or focused position (Rousou, 2010), as in the following:

- (2) I Maria theli [o Yanis na aghapai mono aftin].
The.NOM Maria.NOM want.3SG.PRS the.NOM Yani.NOM SBJV love.3SG only her.ACC
‘Maria wants Yanis to only love her.’

Tsilia, 2023: p.657, ex.7

The NOM DP construction is not puzzling, since the DP is part of the embedded clause. On the contrary, in the ACC DP construction as in (1) the DP, as I will show, is base-generated outside the embedded CP, and has not moved from the lower clause. Speakers vary with respect to whether they find the ACC DP construction grammatical, and only a subset of attitude verbs permits it, although the phenomenon is quite productive for those speakers that allow it. I tested all the examples in this paper with three native speakers of MG who belong to the group of speakers who find the ACC DP construction grammatical. Here is a non-exhaustive list of verbs that can be used in this construction: *thelo* (want), *perimeno* (expect), *pistevo* (believe), *theoro* (consider), *ipologhizo* (estimate), *ksero* (know), *epithimo* (wish), *thimame* (remember) and—at least for Ingria (1981), although the judgment may vary across speakers—*nomizo* (think).² I should note that some of these verbs take subjunctive complements (*na*-clauses), some indicative complements (*oti*-clauses), and some both:³

¹Brackets indicate hypothesized clause boundaries. MG has subject *pro*-drop and free word order, although VSO is considered to be the unmarked word-order (Philippaki-Warbuton, 1982, 1985, 1987; Alexiadou and Anagnostopoulou, 2001).

²Some verbs that cannot be used in this construction are *apodhiknio* (prove), *krino* (judge), *elpizo* (hope):

- (3) O dhikighoros apedhikse /ekrine /ilpize i Eleni
The.NOM lawyer.NOM prove.PST.PFV.3SG judge.PST.PFV.3SG hope.PST.3SG the.NOM Eleni
 /*tin Eleni oti ine athoa.
 the.ACC Eleni COMP be.3SG innocent
‘The lawyer proved/judged/hoped that Eleni is innocent.’

³I follow Philippaki-Warbuton and Veloudis (1984); Philippaki-Warbuton (1987, 1994, 1998); Tsimpli (1990); Giannakidou (1998) in treating *na* as a subjunctive mood marker rather than a complementizer.

- (4) a. Perimename tin Eleni [oti tha eksorjisti *pro*].
 Expect.1PL.PRS the.ACC Eleni.ACC COMP will be-furious.3SG 3SG
 ‘We expected that Eleni will be furious.’

Kotzoglou, 2017: p.310, ex.29, brackets and *pro* are mine

- b. Perimename tin Eleni [na eksorjisti *pro*].
 Expect.1PL.PRS the.ACC Eleni.ACC SBJV be-furious.3SG 3SG
 ‘We expected that Eleni will be furious.’

The ACC DP is semantically related to a (potentially silent) pronoun in the embedded clause. As shown in Tsilia (2023), this pronoun does not have to be in the subject but can also be in the object position. When in subject position, because of *pro*-drop, this pronoun is often a *pro* (unless focused or in a coordinated DP); but when in object position, the pronoun is necessarily overt:

- (5) *Yanis is a political activist and part of an organization run by me. I want to raise awareness about it and I think that getting someone arrested will give us some publicity to this end.*

Thelo **ton** **Yani_i** [na *ton_i* silavi i astinomia].
 Want.PRS the.ACC Yani.ACC SBJV him.ACC arrest the.NOM police.NOM

‘I want the police to arrest Yanis.’

Tsilia, 2023: p.657, ex.8

What about other argument positions? It can also be the second argument of a ditransitive as shown in (7).⁴

- (7) *Yanis is working for a publishing house. Maria is a famous writer and she just sold the rights of her new book to a rival publishing house. I am the boss of Yanis and I send him at a book fair to meet Maria and convince her to transfer to him the rights of her book.*

Thelo **ton** **Yani_i** [na *tu_i* dhosi to vivlio i Maria].
 Want.PRS the.ACC Yani SBJV he.DAT give the.ACC book.ACC the.NOM Maria

‘I want Maria to give the book to Yanis.’

Tsilia, 2023: p.667, ex.36

Yet, the syntactic status of the *na* particle is not crucial for my analysis.

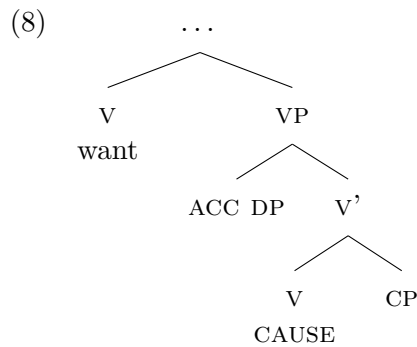
⁴Interestingly, it cannot be in an adjunct position, as in the PP in (6):

- (6) *I want Yanis to work hard, get good grades, and make his mom proud.*

*Thelo **ton** **Yani_i** [na ine perifani i mama *tu_i* jia afton_i].
 Want.PRS the.ACC Yani SBJV be.3SG proud the.NOM mom.NOM his.GEN for him.ACC
 Intended: ‘I want from Yanis to make his mom proud about him.’

Instead, the intended meaning in (6) would be expressed via a PP (i.e., I want *from Yanis*). The restriction of the pronoun to argument positions will be left as an open puzzle.

Building on syntactic literature that argues that the ACC DP is base-generated above the embedded CP, I present an *intensionality paradox*: despite being outside the embedded clause, the ACC DP can have *de dicto*, i.e., low-scope and opaque (Fodor, 1970a), readings. It can be interpreted with respect to the worlds quantified over by the attitude verb, and thus be part of its intensionalized argument. Additionally, there is a *causal semantic requirement* that the referent of the ACC DP must be able to cause the event described by the embedded clause. I provide a clausal analysis, arguing that there are three instead of two clauses in the examples shown above. I argue that the ACC DP is case-marked by the matrix verb but is the subject of an intermediate VP-clause headed by a concealed causative CAUSE. Thus, it is part of the intensionalized argument of the attitude verb without being part of the (most) embedded clause. I argue for the following syntax:



Assuming this syntax, the ACC DP is part of the intensionalized argument of the attitude verb WANT, and thus the *de dicto* readings follow.⁵ This contributes to a broader debate around intensionality, showing that apparent semantic scope paradoxes can be solved if we assume silent predicates, when we are justified to do so.

The rest of this paper is organized as follows. Syntactically, I show that the ACC DP is not part of and has not moved from the *na*- or *oti*-marked embedded clause in Section 2. Semantically, after comparing the construction with prolepsis in Section 3, I present the intensionality paradox, namely that despite not being part of the (most) embedded clause the ACC DP can be read *de dicto*, suggesting it semantically behaves as though in the scope of the attitude verb. Then, in Section 5 I discuss the causal semantics of the construction, proposing a concealed causative. I provide a clausal analysis in Section 6, arguing that there is a covert proleptic CAUSE in between the attitude verb and the ACC DP. I show evidence for such a causative in Section 7. Finally, I compare the construction with other concealed causatives in Section 8, and address some issues in Section 9.

⁵Throughout the paper, I use the verb ‘want’ to illustrate this construction. However, the semantic and syntactic tests of this paper apply to the other verbs participating in the construction as well.

2 Syntactic Position of the ACC DP

The syntax of this construction, and in particular the position of the ACC DP, is debated. There are a number of plausible analyses: (i) it could be Exceptional Case Marking (ECM), where the ACC DP is somewhere inside the CP (Philippaki-Warbuton and Spyropoulos, 1996; Hadjivassiliou et al., 2000; Pratt, 2011), (ii) it could be raising, where the ACC DP is base-generated in the embedded CP and then moves to the matrix clause (Philippaki-Warbuton, 1979; Kakouriotis, 1980), or (iii) it could be that the ACC DP is actually base-generated above the CP and is semantically connected to a (possibly silent) pronoun inside it. There are two versions of the latter hypothesis depending on whether the ACC DP is semantically related only to the subject of the embedded clause, in which case it could be object control (Kotzoglou, 2002; Kotzoglou and Papangeli, 2007; Kotzoglou, 2017), or whether it can also be semantically related to other arguments inside the CP, in which case it could be prolepsis (Tsilia, 2023). I will argue for yet another syntactic representation, where there are three instead of two clauses, with the ACC DP being the subject of a concealed causative (see the tree in (8) above). For that, I need to first establish that the ACC DP (a) is not part of the lower CP, and (b) it did not move from there (and was thus base-generated above it). I will draw on arguments from Kotzoglou (2002); Kotzoglou and Papangeli (2007); Kotzoglou (2017), while also refining them and providing a new argument from the absence of island effects.

Let us start with arguments that the ACC DP is syntactically not part of the lower CP.

2.1 Prepositional Phrases

Kotzoglou (2002) notes that Prepositional Phrases (henceforth PPs) can intervene between the ACC DP and the CP. In this case, there are two available parses for the PP (indicated by brackets): it can either modify the matrix verb as in (9a) or the embedded one as in (9b), where the PP occupies a topicalized or focused position in the left periphery. Crucially, the meaning changes; the habit ends up being the son’s (9a) or the father’s (9b) accordingly.

- (9) a. *Ithele ton patera tu apo sinithia [na tu lei pro kathe*
Want.PST the.ACC father.ACC he.GEN by habit SBJV he.GEN say 3SG every
mera ke ena dhiaforetiko paramithi].
day and a.ACC different.ACC fairy-tale.ACC
‘By habit, he wanted his father to read to him a different fairy tale every day.’
(son’s habit)

- b. Ithele **ton** **patera** **tu** [apo sinithia na tu lei *pro* kathe
 Want.PST the.ACC father.ACC he.GEN by habit SBJV he.GEN say 3SG every
 mera ke ena dhiaforetiko paramithi].
 day and a.ACC different.ACC fairy-tale.ACC
- ‘He wanted his father to read to him by habit a different fairy tale every day.’
 (father’s habit)

Kotzoglou, 2002: p.45, ex.7(b), brackets and *pro* are mine

The PP modifying the matrix verb with the ACC DP on its left in (9a) suggests the ACC DP acts as part of the matrix clause. This does not yet rule out the possibility that it has moved there from the embedded clause.

To complete Kotzoglou’s 2002 argument, I add that when the DP surfaces in NOM, everything on its right has to belong to the embedded clause. Thus, if it precedes the PP, only one interpretation is possible, namely the PP has to modify the embedded verb as in (10):

- (10) Ithele [o pateras tu apo sinithia na tu lei kathe mera
 Want.PST the.NOM father.NOM he.GEN by habit SBJV he.GEN say every day
 ke ena dhiaforetiko paramithi].
 and a.ACC different.ACC fairy-tale.ACC
- ‘He wanted his father to read to him by habit a different fairy tale every day.’
 (father’s habit)

Kotzoglou, 2002: p.46, ex.8(b), brackets are mine

Unlike the ACC DP, the NOM variant and its rightward elements belong to the embedded clause. As a control, notice that if the PP precedes the NOM DP, the sentence is ambiguous between an embedded and a matrix attachment of the PP:

- (11) Ithele apo sinithia o pateras tu na tu lei kathe mera
 Want.PST by habit the.NOM father.NOM he.GEN SBJV he.GEN say every day
 ke ena dhiaforetiko paramithi.
 and a.ACC different.ACC fairy-tale.ACC
- ✓ ‘He wanted his father to read to him by habit a different fairy tale every day.’
 (father’s habit)
- ✓ ‘By habit, he wanted his father to read to him a different fairy tale every day.’
 (son’s habit)

What this shows is that the ACC DP is part of a higher clause, since it can have higher clause material on its right. The contrary holds for the NOM DP, which necessarily marks a clause boundary; it is part of the embedded clause, as is anything on its right.

2.2 Negative Polarity Items

Negative Polarity Item (henceforth NPI) licensing also shows the ACC DP is above the embedded clause. Kotzoglou (2002) shows that when the ACC DP is an NPI, it cannot be licensed by the negation in the embedded CP; but when the NOM DP is an NPI, it is licensed by the CP-internal negation. Here is the minimal pair:⁶

- (12) a. *Me tetia siberifora perimena **KANENA** [na mi me
 With such.ACC behavior.ACC expect.PST nobody.ACC SBJV NEG me.ACC
 proslavi *pro* sti dhulia tu].
 hire.3SG 3SG in-the job he.GEN

Intended: ‘With such a behavior, I expected nobody to hire me at their job.’

Kotzoglou, 2002: p.47, ex.10(b), brackets and *pro* are mine

- b. Me tetia siberifora perimena [KANIS na mi me
 With such.ACC behavior.ACC expect.PST nobody.NOM SBJV NEG me.ACC
 proslavi sti dhulia tu].
 hire.3SG in-the job he.GEN

‘With such a behavior, I expected nobody to hire me at their job.’

Kotzoglou, 2002: p.47, ex.10(a), brackets are mine

The NPI in (12b) is part of the embedded CP, having moved to its left periphery; it can reconstruct for licensing within its clause. However, the NPI in (12a) cannot be licensed in the same way, suggesting that it is not part of the embedded clause. Instead, a higher matrix clause negation is needed to license it:

- (13) Me tetia siberifora dhen perimena **KANENA** [na me proslavi
 With such.ACC behavior.ACC NEG expect.PST nobody.ACC SBJV me.ACC hire.3SG
pro sti dhulia tu].
 3SG in-the job he.GEN

‘With such a behavior, I expected nobody to hire me at their job.’

To complete Kotzoglou’s 2002 argument, I add the following control, showing that an object NPI that has moved to the left periphery in the embedded CP can reconstruct for licensing:

- (14) Me tetia chrimatodhotisi perimena [KANENA_i na mi proslavun *pro*
 With such.ACC funding.ACC expect.PST nobody.ACC SBJV NEG hire.3PL 3PL
t_i].
t

‘With such funding, I expected them to hire nobody.’

⁶The NOM DP nobody can also be pronounced KANENAS in (12b).

So, the fact that (12a) does not reconstruct shows us that it is not part of the embedded clause. Indeed, [Tsimpli and Roussou \(1996\)](#) also show that an NPI can reconstruct for licensing within its clause:

- (15) Apofasisa [TIPOTA na mi fao t].
 Decide.PST.1SG nothing SBJV NEG eat.1SG t
 ‘I decided not to eat anything.’

[Tsimpli and Roussou, 1996](#): p.59, ex.26(b)

However, notice that we have not yet excluded the possibility that the ACC DP moved into the matrix clause. Indeed, [Tsimpli and Roussou \(1996\)](#) provide the following example, where an NPI moves into the matrix clause and fails to be licensed:

- (16) *TIPOTA apofasisa [na mi fao t].
 nothing decide.PST.1SG SBJV NEG eat t
 Intended: ‘I decided not to eat anything.’

[Tsimpli and Roussou, 1996](#): p.59, ex.24(b)

Thus, it seems that although an NPI can reconstruct for licensing within its clause, it cannot do so if it moved to a different clause.⁷

Overall, the NPI data show us that the ACC DP is part of a clause higher than the embedded one. Since it cannot be licensed by a negation in the embedded clause and given that NPis do reconstruct for licensing within their clause, we can conclude that the ACC DP, contrary to the NOM one, is part of a higher clause. Indeed, had it been part of the embedded clause as in (12b) or (14), it would have reconstructed for licensing. Therefore, so far we have seen that the ACC DP is part of a higher clause, without yet excluding the possibility that it was base-generated in the embedded clause and then moved to the higher clause.

2.3 Clitic Doubling

The ACC DP behaves like an argument of the higher verb in that it can be clitic doubled (see also [Kotzoglou and Papangeli, 2007](#): p.123, ex.29(b)):

⁷Although [Kotzoglou and Papangeli \(2007\)](#) provide the following example, where the NPI *kanis* is licensed by the negation in the embedded clause and then moves to the matrix clause:

- (17) KANIS nomize i Fani [oti dhen aghorase efimeridha].
 nobody.NOM think.PST.3SG DET Fani.NOM COMP NEG buy.PST.3SG newspaper.ACC
 ‘Fani thought that nobody bought a newspaper.’

[Kotzoglou and Papangeli, 2007](#): p.121, ex.24(a)

There could be cross-speaker variation or a difference between *na*- and *oti*-clauses. For the purposes of this paper, I leave this open for future research, taking the conservative route of assuming that NPis cannot be licensed in their clause and then move out of it.

- (18) **ton_i** perimena **ton** **Yani_i** [na aghapisi *pro* ti Maria].
 CL.ACC expect.PST.1SG the.ACC Yani.ACC SBJV love.3SG 3SG the.ACC Maria.ACC
 ‘I expected Yanis will love Maria.’

As a control, notice that we cannot clitic double a NOM DP in the matrix clause:

- (19) ***ton_i** perimena [o Yianis_i na aghapisi ti Maria].
 CL.ACC expect.PST.1SG the.NOM Yani.NOM SBJV love.3SG the.ACC Maria.ACC
 Intended: ‘I expected Yanis will love Maria.’

This shows that the ACC DP behaves as an argument of the higher clause.

2.4 Absence of reconstruction

I have by now established that the ACC DP is not part of the embedded clause. I will now argue that it has not moved from it either.

This can be shown by the absence of reconstruction, since, had it raised, it would be able to reconstruct whenever other moved elements are able to. [Spyropoulos and Revithiadou \(2007\)](#); [Kotzoglou \(2017\)](#) notes that inverse scope interpretations are possible with quantificational subjects or objects:

- (20) enas ipurghos episkeftotan kathe periochi tis Eladhas
 a.NOM minister.NOM visit.PST.3SG every region the.GEN Greece.GEN
 ‘A minister was visiting every region of Greece.’

[Spyropoulos and Revithiadou, 2007](#): p.6, ex,11(b)

The above sentence indeed has both surface scope (i.e., there is a minister that was visiting every city) and inverse scope (i.e., for every city, there is a minister that is visiting it). Similarly, when we have a quantificational expression in the NOM DP position two readings are available:

- (21) sto parti perimena [enas filos mu na fai kathe
 at-the party expect.PST.1SG a.NOM friend.NOM mine.GEN SBJV eat every
 tiropitaki].
 cheese-pie.DIM.ACC
 ‘At the party, I expected a friend of mine to eat every cheese pie.’
 ✓Context A, ✓Context B

This is compatible with the following contexts:

- (22) a. Context A: My wife and I threw a party, where I invited some of my friends and she invited some of hers. Her friends were all vegan so they couldn’t eat the cheese

pies. But all my friends love eating cheese pies, so I expected they would eat all of them.

- b. Context B: My wife and I threw a party, where I invited some of my friends and she invited some of hers. A friend of mine, George, loves cheese pies so much so I expected he would end up eating all of them.

However, no inverse scope readings are possible when the quantificational object is in the ACC DP position:

- (23) *sto parti perimena enan filo mu [na fai pro kathe*
 at-the party expect.PST.1SG a.ACC friend.ACC mine.GEN SBJV eat.3SG 3SG every
tiropitaki].
 cheese-pie.DIM.ACC

‘At the party, I expected a friend of mine to eat every cheese pie.’

✗Context A, ✓Context B

[Kotzoglou, 2017](#): p.306, ex.17(a), contexts and *pro* are mine

(23) only has a surface scope interpretation (i.e., there is a particular friend whom I expected to eat everything). By contrast, in (21) inverse scope is also possible.

What does the absence of inverse scope in (23) tell us? Inverse scope is the result of (i) reconstruction of the surface-higher quantifier to a lower copy position, or (ii) Quantifier Raising (QR). The fact that the inverse scope reading is inaccessible here shows that neither option is available. Given that the ACC DP is not inside the CP, QR of the embedded quantificational expression across it would be ruled out by locality conditions. But why is reconstruction of the ACC DP into a lower copy position not available here? There are two possibilities. First, the absence of reconstruction could suggest the absence of a trace position in the CP where the ACC DP could have moved from. In turn, if the ACC DP did not move from the lower clause and given that I have argued that it is part of a higher clause, it must have been base-generated at the higher clause that it is also part of. The other possibility is that the ACC DP moved from the lower clause but cannot reconstruct for scope because of the clause boundary. In what follows, I provide further evidence against this second option, and I argue that there is in fact no movement from the lower clause.

2.5 Coordinated DP island

This subsection provides a new argument against a movement approach. In the examples above with an ACC DP, there is a gap inside the complement clause. I suggest that this gap is a silent pronoun (MG being a *pro*-drop language). This pronoun linked to the ACC DP can be inside a coordinated DP island, and in this case, the pronoun is overt:

- (24) *Maria's dad usually does not like to meet her boyfriends, but yesterday he was drunk and he wanted Maria and her boyfriend to come for dinner some day. Today, he sobered up and changed his mind again.*

- a. O babas tis ithele chtes **ti** **Maria_i** [na
 The.NOM dad.NOM hers want.PST yesterday the.ACC Maria.ACC SBJV
 erthi **afti_i** kai to aghori tis jia fajito mia mera].
 come.3SG she.NOM and the.NOM boy.NOM hers for food one day
 'Her dad wanted yesterday Maria and her boyfriend to come for dinner one day.'
- b. *O babas tis ithele chtes [i Maria_i na erthi
 The.NOM dad.NOM hers want.PST yesterday the.NOM Maria.NOM SBJV come
afti_i kai to aghori tis jia fajito mia mera].
 she.NOM and the.NOM boy.NOM hers for food one day

The DP is grammatical in the ACC, but not in the NOM.⁸ This test shows that the ACC DP cannot have moved from the lower clause, since the pronoun in the CP is in an island environment out of which it cannot move/raise. Conversely, the NOM DP, which would need to move out of the coordinated DP island to SpecCP, is ungrammatical, precisely because it cannot move out of an island environment. The ACC DP is grammatical when movement is impossible because it never moves from the lower clause. It is base-generated in the higher clause and is only semantically related to the pronoun in the CP.

2.6 Idiomatic Readings

Yet another argument against a raising analysis comes from idiomatic readings. Had the ACC DP been raised, it could be part of an idiom in the embedded clause (since there would be no extra theta role assignment), as in the following English examples:

- (25) a. The shit has hit the fan.
 b. I expected the shit to have hit the fan.

But, if there is no movement of the ACC DP and there are rather two syntactically independent clauses, we would expect idioms to be ungrammatical, as in control cases:

- (26) *I persuaded the shit to hit the fan.

Indeed, the construction in question in MG patterns with control in this case, as shown by the following data from [Kotzoglou and Papangeli \(2007\)](#):

⁸Note that the temporal adverb *chtes* 'yesterday' was included here to better reflect the context, to separate the two clauses as well as the time of the desire from the time of the meal. That being said, the adverb is not crucial for the judgment, and could be omitted or surface to the right of the ACC DP.

- (27) a. perimena **#psilus** / [psili na tu bun st' aftia].
 expect.PST.1SG fleas.ACC fleas.NOM SBJV he.GEN get.3PL in-the ears
 'I expected fleas to get into his ears (I expected him to become suspicious).'

Kotzoglou and Papangeli, 2007: p.124, ex.33(a)

- b. me afta pu eleghe perimena **#ton** **dhjaolo** / [o
 with these COMP say.PST.3SG expect.PST.1SG the.ACC devil.ACC the.NOM
 dhjaolos na ton pari].
 devil.NOM SBJV him take.3SG

'With the things he said I expected the devil to take him (i.e., him to be destroyed).'

Kotzoglou and Papangeli, 2007: p.125, ex.33(b)

The unavailability of the idiomatic reading suggests that the matrix clause and the CP are syntactically independent and that the ACC DP is not a raised object, but rather a base-generated one.

2.7 Not object control

So far, I have argued (a) that the ACC DP is syntactically part of a higher clause (see PPs, clitic doubling and NPI behavior), and (b) that it was base-generated there (on the basis of the absence of reconstruction effects, insensitivity to islands, unavailability of idiomatic readings). This excludes an ECM as well as a raising analysis, leaving us with the option between object control, prolepsis, or a concealed causative analysis.

As shown in Tsilia (2023) (see (5)), the construction in question is not object control, since given the right context, the pronoun can be in object position too.⁹ This leaves us the choice between a proleptic analysis (Tsilia, 2023) and the concealed causative analysis I will propose here. In the next section, I compare the construction with prolepsis cross-linguistically,

⁹Another argument against an object control analysis comes from *wh*-extraction. Hadjivassiliou et al., 2000: p.76, ex.17 and Kotzoglou and Papangeli, 2007: p.129, ex.42 mention that *wh*-extraction is banned in the construction in question, as opposed to object control:

- (28) a. ??pjon itheles ton prothipurgho [na entiposiasi *pro t_{wh}*] ?
 who.ACC want.PST the.ACC prime-minister.ACC SBJV impress.3SG 3SG
 Intended: 'Who did you want the prime minister to impress?'

Kotzoglou and Papangeli, 2007: p.129, ex.42(a)

- b. pjon epises ton prothipurgho na entiposiasi?
 who.ACC persuade.PST the prime-minister.ACC SBJV impress.3SG
 'Who did you persuade the prime minister to impress?'

Kotzoglou and Papangeli, 2007: p.129, ex.42(b)

Notice that extraction becomes even worse if a clitic is added (first noted in Hadjivassiliou et al., 2000: p.77, ex.18(c) and 19(c)), as in the following:

- (29) *pjon ton itheles ton prothipurgho [na entiposiasi *pro t_{wh}*] ?
 who.ACC him.ACC want.PST the.ACC prime-minister.ACC SBJV impress.3SG 3SG
 Intended: 'Who did you want the prime minister to impress?'

As a control, note that we can otherwise clitic double a *wh*-item in Modern Greek:

arguing that while it is a promising analysis, there are some puzzling differences between the MG construction and prolepsis in other languages.

3 Properties of prolepsis

There are many parallels between the construction in question and prolepsis, and such an analysis was in fact pursued in Tsilia (2023). Prolepsis, as Salzmann (2017a: p.1) defines it is “a construction where a structural complement of the matrix verb is semantically related to the predicate of the embedded clause without there being an obvious movement relationship”. Cross-linguistically, proleptic constructions have a limited range of interpretations. The proleptic object needs to be specific in many languages, i.e. if it is an indefinite it needs to scope above the attitudinal quantifier. It also needs to be read transparently, i.e., evaluated with respect to the actual world.

For example, in the German example below, the pronoun *es* ‘her’ in the CP is semantically related to *einem Mädchen* ‘a girl’, despite the fact that the latter is base-generated in the matrix clause:

- (32) **Von einem Mädchen** weiss ich, dass Peter es geküsst hat.
 Of a-DAT girl know-1SG-PRES I COMP Peter her kiss-PTCP have-3SG
 ‘Of a girl, I know that Peter kissed her.’

Salzmann, 2017a: p.15, ex.30(a)

This sentence only has a *de re* reading, i.e. a specific and transparent reading. This means that there is a specific girl that the speaker knows Peter kissed, and this girl exists in the actual world.

In Tiwa (Dawson and Deal, 2019), such constructions also have the so-called third reading. The NP-complement is still evaluated transparently with respect to the actual world, but the speaker need not have a specific individual in mind. In the following example, Mukton does not think that a specific mosquito will bite him, but rather that some mosquito will do so:¹⁰

- (30) pjon ton itheles t_{wh} ?
 who.ACC him.ACC want.PST.2SG
 ‘Who did you want?’

Finally, by contrast to (28a), *wh*-extraction is perfectly possible if the DP is in the NOM:

- (31) pjon itheles [o prothipurghos na entiposiasi t_{wh}]?
 who.ACC want.PST the.NOM prime-minister.NOM SBJV impress
 ‘Who did you want the prime minister to impress?’

¹⁰The verb used in the Tiwa examples here is *think*, but Dawson and Deal (2019) notes that the process is very productive, extending to most attitude verbs. A minimal comparison can be made between Greek and Tiwa with the verbs *know*, *believe*, and *remember*:

(35) *Mukton does not want to go outside.*

pro_j **kishá khódo-gô_i** atkhâl lá-ga, [*pro_i* pe-go_j chi-w honmandé]
 3SG one mosquito-ACC think PFV 3SG 3SG-ACC bite-NEUT COMP

‘He thinks a mosquito will bite him.’

Dawson and Deal, 2019: p.330, ex.3

In the example above, there does not have to be a specific mosquito Mukton thinks will bite him. However, there need to be mosquitoes in the actual world; the mosquitoes cannot just exist in Mukton’s belief worlds. This combination of a non-specific and yet transparent reading is what makes this a third reading. In other words, the NP-complement is evaluated with respect to the actual world, but the speaker does not have a specific mosquito in mind. Crucially, however, in Tiwa just like in other instances of prolepsis, *de dicto* readings are unavailable as Dawson and Deal (2019) show:

(36) *Tonbor is not very smart. He doesn’t know that dogs can’t be green.*

#Tonbor **kishá khó dang-shór khú gri-gô_i** atkhâl lá-ga, [Lastoi *pro_i* pre-ga
 Tonbor one green dog-ACC think-PFV Lastoi 3SG buy-PFV
 honmandé]
 COMP

Intended: ‘Tonbor thinks that Lastoi bought a green dog.’

Dawson and Deal, 2019: p.338, ex.35

The above example is infelicitous, since it commits the speaker to the existence of green dogs in the actual world. The DP *kishá khó dang-shór khú gri-gô* ‘a green dog’ cannot be read opaquely to put it in Fodor’s terms (Fodor, 1970a), i.e., it cannot be interpreted in the worlds quantified over by the attitude verb—the worlds compatible with Tonbor’s thoughts. Instead, it can only have a transparent reading, according to which green dogs exist in the actual world.

What is common to proleptic constructions described in the literature so far (Davies, 2005; Salzmann, 2017a,b; Deal, 2018; Dawson and Deal, 2019) is the fact that the *de dicto* reading of the proleptic nominal is impossible. This justifies, in a sense, a specific construction that marks *de re* (in the case of German (Salzmann, 2017a,b) and Nez Perce (Deal, 2018) or non-*de dicto* (in the case of Tiwa (Dawson and Deal, 2019)). Given that the proleptic objects have not moved out of the embedded CP, they have to be interpreted transparently, with respect to the world of evaluation. This is because they are scopally outside the intensionalized

(33) Sonali **Mansing-go_i** si-ga/ nol-ga/ khósoi mán-ga, [*pro_i* lí-ga honmandé]
 Sonali Mansing-ACC know-PFV/ believe-PFV/ remember-PFV 3SG go-PFV COMP
 ‘Sonali knows/believes/remembered that Mansing went.’

(34) I Maria kseri/ pistevi/ thimate **ton Yani_i** [oti efije *pro_i*]
 DET Maria know.PRS.3SG/ believe.PRS.3SG/ remember.PRS.3SG DET Yani.ACC COMP left.3SG 3SG
 ‘Maria knows/believes/remembers of Yanis that he left.’

Virginia Dawson, p.c.

argument of the verb, i.e. the embedded CP, and thus cannot be evaluated with respect to the worlds quantified over by the attitude verb. Below is a table summarizing the empirical picture for prolepsis:

Language	<i>De re</i>	Third	<i>De dicto</i>
	specific transparent	non-specific transparent	non-specific opaque
Madurese, German, Nez Perce	✓	✗	✗
Tiwa	✓	✓	✗

Table 1: Readings of prolepsis cross-linguistically

[Tsilia \(2023\)](#) argues in favor of a proleptic analysis of the MG construction, since the ACC DP is semantically related to a pronoun in the embedded CP and yet has not moved from it. However, there is an important difference between the MG construction and proleptic constructions in other languages. In MG, the ACC DP can also have *de dicto* readings, as I will show in the next Section. In other words, despite being outside the embedded CP, the ACC DP can be interpreted in a non-specific and opaque way, as if it is part of the intensionalized argument of the attitude verb. To account for this, [Tsilia \(2023\)](#) assumes a semantic lowering mechanism. In this paper, I will argue that such a mechanism is not needed, and I will dispense with the proleptic lexical entry for *thelo* ‘want’ proposed in that work. I will instead take *de dicto* readings, as well as a causal semantic requirement associated with the ACC DP, to indicate that it is part of an intermediate clause containing a concealed causative without being part of the (most) embedded CP.

4 Is the ACC DP in the scope of the attitude verb?

The discussion of proleptic objects in the previous Section suggests that proleptic objects are semantically read transparently *because* they cannot reconstruct within the scope of the attitude verb. More specifically, it seems that semantically the NP-complement of a proleptic object in a given language cannot be evaluated with respect to the worlds quantified over by the attitude verb because syntactically there is no position inside the embedded clause where the NP (or a DP containing it) can reconstruct.

However, in the case of MG the syntax and the semantics seem to diverge. Syntactically, we have established that the ACC DP is not part of the lower CP. This might lead us to expect that its NP-complement has to be read transparently like in the cases of prolepsis above. Yet, I will show that this is not the case. Semantically, it can be read opaquely, suggesting that it is in the scope of the attitude verb. The puzzle will then be: where is this mismatch at the syntax-semantics interface stemming from? Is the NP-complement of the ACC DP in or out of the scope of the attitude verb?

Tsilia (2023) already showed that the ACC DP can have *de dicto* readings, being read opaquely:

- (37) *Little Petros is in kindergarten and he and his friends believe that green dogs exist. One day they are talking about green dogs and Petros bets that exactly three of them will show up at his party.*

O Petrakis theli akrivos **tris** **prasinus** **skilus_i**
 The.NOM Petros.DIM.NOM want.PRS exactly three.PL green.ACC.PL dog.ACC.PL
 [na erthun *pro_i* sto parti].
 SBJV come.3PL 3PL in-the party

‘Little Petros wants three green dogs to come to the party.’

Tsilia, 2023: p.665, ex.30, *pro* mine

This attitude report does not commit the speaker to the existence of green dogs; the embedded subject is read opaquely. What is more, in cases of mistaken identity, it is felicitous to use the ACC DP, while asserting that its denotation is empty in the actual world. This is the case in the following scenario:

- (38) *This morning I was moving into my new apartment and had lots of moving boxes. A woman who was helping me move the boxes saw a tenant of the building looking a lot like me and she mistakenly thought it’s my brother. She did not know that in reality I have no brother. So, this woman seeing that tenant and thinking it’s my brother, she wanted to ask him to help us with the boxes.*

I kiria pu me voithuse ithele **ton** **aderfo**
 The.NOM lady.NOM who me.ACC help.IMP.F.PST want.PST the.ACC brother.ACC
mu_i [na mas voithisi *pro_i*], ala eghe dhen echo adherfo!
 mine.GEN SBJV us.ACC help.3SG 3SG but I NEG have.PRS brother

‘The woman helping me wanted my brother to help us, but I do not have a brother.’

There are also non-specific readings, as in the following:

- (39) I Zoi theli **mian nosokoma_i** [na ti voithisi *pro_i*].
 The.NOM Zoi.NOM want.PRS a.ACC nurse.ACC SBJV her.ACC help.3SG 3SG.
 Opiadhipote nosokoma.
 Any.FEM nurse.

‘Zoi wants a nurse to help her. Any nurse.’

The example above shows that *mian nosokoma* ‘a nurse’ is read non-specifically, since it does not matter for Zoi’s desires which nurse actually helps Zoi. Similarly, in the following example *enan sinadhelfo* ‘a colleague’ is read non-specifically, since it does not matter for Stergios’ desires who helps him:

- (40) *Stergios is desperately looking for a colleague to help him finish his task before this evening. The task does not require any expertise so any colleague would do.*

O Stergios theli apla **enan sinadhelfo tu_i** [na
 The.NOM Stergios.NOM want.PRS simply a.ACC colleague.ACC his.GEN SBJV
 ton voithisi *pro_i*]. Dhen echi simasia pios / #pion.
 him.ACC help.3SG 3SG NEG has importance who.NOM who.ACC.

‘Stergios simply wants a colleague of his to help him.’

✓‘It does not matter who [helps him]’

✗‘It does not matter whom [he wants to help him]’

The only felicitous reading this sentence has is one in which Stergios wants some colleague to help him and it does not matter to him who ends up helping him. There is no reading in the context according to which Stergios wants a specific colleague to help him (e.g., John) but it does not matter who he has in mind. This further shows the existence of a non-specific reading.

Finally, the ACC DP’s NP-restrictor can be empty, as we can see in the following examples, given that there are no green dogs or dragons in the actual world:

- (41) a. O Petrakis theli **prasinus skilus_i** [na erthun *pro_i*
 The.NOM Petros.DIM.NOM want.PRS green.ACC dogs.ACC SBJV come.3PL 3PL
 sto parti tu].
 to-the party his.GEN

‘Little Petros wants green dogs to come to his party.’

- b. O Yanis o elafroiskiotos theli **enan dhrako_i**
 The.NOM Yanis.NOM the.NOM moonstruck want.PRS.3SG a.ACC dragon.ACC
 [na epitethi *pro_i* ston astinomiko].
 SBJV attack.3SG 3SG to-the policeman

‘Yanis the moonstruck wants a dragon to attack the policeman.’

Apart from *de dicto* readings, classic *de re* readings are, as expected, also permitted:

- (42) *Maria is on an apostolic mission in Egypt during the pandemic, while working remotely. She started this job during COVID and thus never got to meet her colleagues, Markos and Christos. It just so happens that Markos and Christos are also in Egypt and Maria has met them without knowing they are her colleagues. She tried to convince them to become Catholic.*

I Maria theli **kathe** **tis** **sinadhelfo**_i [na ine *pro*_i
The.NOM Maria want.PRS every.ACC she.GEN colleague.ACC SBJV be.3SG 3SG
katholikos].
catholic.NOM

‘Maria wants every colleague of hers to be Catholic.’

[Tsilia, 2023](#): p.665, ex.31, *pro* mine

The ACC DP in MG may also have non-specific but transparent, i.e., third, readings (unsurprisingly so, since they are a combination of *de dicto* and *de re*):

- (43) *Zoi is attending a 100m race at the Olympics. Three contestants are talking to each other before the start. Unbeknownst to Zoi, these three contestants are my friends. She thinks to herself that she wants one of those three people to win the race, because they seem motivated.*

I Zoi theli [enan filo mu]_i [na kerdhisi *pro*_i ton
The.NOM Zoi want.PRS a.ACC friend.ACC mine SBJV win.3SG 3SG the.ACC
aghona].
race.ACC

‘Zoi wants a friend of mine to win the race.’

[Tsilia, 2023](#): p.665, ex.32, *pro* mine

In the above example, Zoi does not have any desires about specific friends of mine, and yet the ACC DP’s NP-complement is evaluated with respect to the actual world, not Zoi’s desire worlds.

The ACC DP thus semantically behaves as if it is part of the intensionalized argument of the attitude verb, since it can be read opaquely. The NP predicate can be empty in the actual world, being interpreted solely in the worlds quantified over by the attitude verb, i.e., the ones compatible with the attitude holder’s desires. Thus, we have a mismatch at the syntax-semantics interface, with the ACC DP being syntactically not part of the lower clause, and yet semantically behaving as being in the scope of the attitude verb. What is more, because it can be read opaquely, it does not make a good candidate for a proleptic object in the sense explored in the previous literature. The question then becomes: if the ACC DP is not part of the lower clause, but is part of the intensionalized argument of the attitude verb, what exactly *is* in the scope of the attitude verb? I call the fact that the lower clause does not contain the ACC DP, and yet it can be read opaquely, an apparent *intensionality paradox*. I will propose a clausal analysis, solving this paradox by arguing that there is in fact an intermediate clause which is the intensionalized argument of the attitude verb. This intermediate clause contains a concealed causative CAUSE, which is motivated by the causative semantics of the construction, elaborated on in the next Section.

5 Causal semantic requirement

An intuition present in much of the earlier work on this construction (Hadjivassiliou et al., 2000; Kotzoglou, 2002; Kotzoglou and Papangeli, 2007; Tsilia, 2023) is that using the ACC DP as opposed to the NOM one gives rise to a different meaning. Various authors formulate this intuition differently. For example, Hadjivassiliou et al. (2000) note that when there is a NOM DP we are expecting an event, but when there is an ACC DP we expect the referent to perform the event described by the embedded clause. Kotzoglou (2002) observes that the same referent is the person or the thing that needs to undertake the action described by the embedded verb. In an attempt to account for this intuition, Kotzoglou and Papangeli (2007) argue that the ACC DP receives a weak theta role, such as “as for...DP” or “on behalf of...DP” from the matrix verb. Tsilia, 2023: p.666-8 argues that there is a ‘causal role requirement’ on the ACC DP, which has to play a salient causal role in realizing the eventuality described by the embedded clause.

The overarching intuition behind these descriptions is that the construction in question is infelicitous when the referent of the ACC DP cannot take some kind of action to cause the event described by the embedded clause. But what exactly is the semantic requirement that the ACC DP has to satisfy in order to be felicitous in this position? This could be a clue to solving the intensionality paradox formulated in the previous Section. Indeed, I will argue that the construction with the ACC DP has causal semantics. More specifically, I will model this with a third, intermediate clause with a concealed causative intervening between the matrix verb and the embedded clause. I will argue that the ACC DP’s referent needs to be capable of causing the eventuality described by the embedded clause, and that the covert CAUSE asserts that it does cause it.

We have already seen that the ACC DP can be semantically related to arguments other than the subject of the embedded clause (see (5)), so the causal semantic requirement cannot be tied to subjecthood. Indeed, subjecthood is not a necessary licensing condition, and it is not sufficient either, since in passive constructions, where subjects are patients, the ACC DP is infelicitous out-of-the-blue:

- (44) a. Thelo **#ton** **Yani** / [o Yanis na silifthi].
Want.PRS the.ACC Yanis.ACC / the.NOM Yanis.NOM SBJV arrest.PASS
‘I want Yanis to be arrested.’

Tsilia, 2023: p.666, ex.33(a)

b. Thelo **tin** **astinomia** / [i astinomia na silavi ton
 Want.PRS the.ACC police.ACC / the.NOM police SBJV arrest the.ACC
 Yani].
 Yani.ACC

‘I want the police to arrest Yanis.’

[Tsilia, 2023](#): p.666, ex.33(b)

Intuitively, (44a) is infelicitous out-of-the-blue because it is not up to Yanis to get arrested; it is the police that are doing so. The infelicity seems to be stemming from the fact that Yanis cannot in principle cause his own arrest. On the contrary, in (44b) the police actively cause the arrest. Interestingly, (44a) is felicitous if the ACC DP’s referent can contextually cause its arrest:

- (45) *Yanis is a political activist and part of an organization run by me. I want to raise awareness about the organization and I think that getting someone arrested will give us some publicity to this end.*

Thelo **ton** **Yani** [na silifthi *pro*]. Tha ine kalo jia tin
 Want.PRS the.ACC Yani.ACC SBJV arrest.PASS.3SG 3SG. Will be good for the.ACC
 kampania.
 campaign

‘I want Yanis to be arrested. It will be good for the campaign.’

[Tsilia, 2023](#): p.667, ex.34

The structure is felicitous only if Yanis is in principle capable of causing the event expressed by the embedded clause. This what I call the *causal semantic requirement* imposed on the ACC DP.

Is this requirement a consequence of the ACC DP receiving an agent theta role from the matrix verb ([Hadjivassiliou et al., 2000](#); [Kotzoglou, 2002](#); [Kotzoglou and Papangeli, 2007](#))? I argue against this, since there is no animacy restriction on the ACC DP.

In the following, the law is in principle capable of causing an eventuality and yet is clearly inanimate:

- (46) *A group of parliamenters is designing a new law about auctions. While brainstorming each parliamenter says what they want out of the new law. One of them says:*

Thelo **ton** **nomo** [na apaghorevi *pro* tis dimoprasies].
 Want.PRS the.ACC law.ACC SBJV prohibit.3SG 3SG the.ACC auctions.ACC

‘I want the law to prohibit auctions.’

Thus, there is no animacy restriction on the ACC DP. All in all, the additional semantic requirement associated with the ACC DP cannot be an agent theta role, since (i) the ACC DP can be inanimate (e.g., (46)), and (ii) it can be a patient in a passive construction as

long as it plays some salient causal role in the eventuality described by the embedded clause (e.g., (45)).

Let us turn to the hypothesis that there is a causal semantic requirement on the ACC DP. If the causal semantic requirement is a necessary condition for the ACC DP to be felicitous, this would account for a puzzle left open by Ingria (1981), concerning the following sentence:

- (47) #Thelo **ton** **Yani** [na tu dhosi to vivlio i Maria].
 Want.PRS the.ACC Yani SBJV he.DAT give the.ACC book.ACC the.NOM Maria

Intended: ‘I want Maria to give the book to Yanis.’

Ingria, 1981: p.203, ex.35

Ingria (1981) presents the oddness of (47) as a puzzle. However, this oddness is expected if there is a causal semantic requirement, since in an out-of-the-blue scenario we do not expect Yanis to cause Maria to give him a book. The sentence becomes felicitous in a scenario where Yanis is able of (partially) causing the eventuality of the embedded clause:

- (48) *Yanis is working for a publishing house. Maria is a famous writer and she just sold the rights of her new book to a rival publishing house. I am the boss of Yanis and I send him at a book fair to meet Maria and convince her to transfer to him the rights of her book.*

Thelo **ton** **Yani** [na tu dhosi to vivlio i Maria].
 Want.PRS the.ACC Yani SBJV he.DAT give the.ACC book.ACC the.NOM Maria

‘I want Maria to give the book to Yanis.’

Tsilia, 2023: p.667, ex.36

The fact that we can manipulate the context to make the ACC DP felicitous is very suggestive. Indeed, if we look at the other felicitous examples, we detect the same causal requirement. In (44b) the police have to act to arrest Yanis, in (45) Yanis has to provoke the police to arrest him, in (48) he has to convince Maria to give him the rights of her book, and in (46) the law has to prohibit auctions. The construction is felicitous only if the ACC DP’s referent can cause the embedded eventuality. For this reason, I suggest that the right generalization capturing the data is that there is a *causal semantic requirement*, stated in the following way:

- (49) Causal semantic requirement: the referent of the ACC DP has to be in principle capable of (at least partially) causing the eventuality described by the embedded clause.

The hidden clause analysis I will propose in the next Section successfully captures this. Contrary to Tsilia (2023), I do not assume a specific lexical entry for a causal proleptic *want*; instead, I argue that the causal semantic requirement follows from a silent CAUSE intervening between the matrix verb and the embedded clause.

The semantics of causatives has been extensively studied in philosophy and linguistics

(Fodor 1970b; Lewis 1973; Dowty 1979; Bittner 1998; Kratzer 2005; Wolff 2007; Copley and Wolff 2014 a.o.). A semantic primitive, which I will notate *cause* with italics (not to be confused with the overt English verb ‘cause’), is frequently invoked to account for the behavior of causative verbs (e.g., Shibatani 1976; Talmy 1976, 1985; Vendler 1984; Van Voorst 1995; Harley 1995; Folli and Harley 2005, 2007; Pylkkänen 2008; Martin 2018 a.o.). Whether *cause* itself constitutes a true semantic primitive or is further decomposable is an interesting question at the interface of philosophy and semantics (cf. Copley and Wolff (2014)), but this issue lies beyond the scope of the present paper. For the purposes of this analysis, I adopt a semantic primitive *cause* to model the MG construction under investigation.¹¹

In many frameworks, *cause* takes two arguments: the causer (an individual or event) and the caused event, i.e., the resultant eventuality of the causation (e.g., Shibatani 1976; Vendler 1984; Parsons 1990; Bittner 1998; Doron 2003; Pylkkänen 2008; Martin 2018 and many others).¹² Here, I treat *cause* as a relation between a causer (an individual) and an event (cf. McCawley 1968, 1971; Vendler 1984; Doron 2003). I will abbreviate this relation as *cause*(*x*, *e*) standing for ‘individual *x* causes event *e*’.¹³

With this background established, I now propose a lexical entry for a covert proleptic CAUSE in Modern Greek, which takes the ACC DP and the embedded clause as its arguments and contains the semantic primitive *cause* in its denotation. In the entry below, which I will elaborate on in the next Section, CAUSE takes two arguments, an embedded predicative CP (Salzmann, 2017a) of type $\langle e, \langle v, \langle s, t \rangle \rangle \rangle$ (taking an individual and an event, and giving a proposition), and the referent of the ACC DP, of type *e*. The causal semantic requirement in (49) above is modeled as a presupposition that the referent of the ACC DP is able to cause the eventuality described by the embedded clause.¹⁴

$$(50) \quad \llbracket \text{CAUSE} \rrbracket = \lambda P_{\langle e, \langle v, \langle s, t \rangle \rangle \rangle} \cdot \lambda x_e \cdot \lambda w_s \cdot \exists e' \exists w' : P(x)(e')(w') \ \& \ \text{cause}(x, e'). \ \exists e[P(x)(e)(w) \ \& \ \text{cause}(x, e)]$$

As mentioned above, *cause* in the right hand side of the above lexical entry is treated as a semantic primitive, denoting a relation between an individual (the causer) and an event (the causee). What the entry above does is presuppose that there is some *P* event in some world caused by the subject; thus, the subject is able of causing the eventuality described by the embedded clause. There are, thus, two components of this lexical entry that require justification: (a) the presupposition, and (b) the assertion. More specifically, one might wonder why the requirement that the ACC DP is capable of causing the eventuality of the

¹¹It is worth noting that the proposed analysis remains compatible with approaches that decompose *cause* as long as the causative meaning can be derived. Also, note that a covert *cause* has also been employed to analyze anticausatives in MG (Alexiadou et al., 2006).

¹²In Dowty (1979), *cause* denotes a relation between propositions.

¹³While this choice impacts the semantic types at the level of the Logical Form, it is not foundational to the core claim, namely, that the MG construction involves a covert causative element between the ACC DP and the embedded clause. This central insight is in principle implementable within alternative theories of causation as well.

¹⁴Throughout the paper, I mark presuppositional material with a colon (:).

embedded clause is a presupposition rather than merely an entailment of the MG concealed causative.

The core of my analysis does not depend on the existence of this presupposition, but rather on the existence of a concealed causative more broadly. That being said, there are some arguments suggesting that the causal semantic requirement is best modeled in terms of a presupposition. As shown in the beginning of this Section, the construction is not always felicitous. More specifically, it is infelicitous in contexts where the referent of the ACC DP is in principle unable of causing the eventuality described by the embedded clause. This is what I called in (49) the causal semantic requirement associated with the construction.

I argue that this requirement is best modeled as a presupposition, since it projects past negation if we negate (5) for example:

- (51) *Yanis is a political activist and part of an organization run by me. I know that Yanis can provoke the police to arrest him, which might get us some media visibility, but this time I want the protest to be peaceful. So, I say:*

Dhen thelo **ton** **Yani_i** [na *ton_i* silavi i astinomia].
 NEG want.PRS the.ACC Yani.ACC SBJV *him*.ACC arrest the.NOM police.NOM

‘I don’t want the police to arrest Yanis.’

The negation in the example above does not target the inference that Yanis is able of causing the police to arrest him. Instead, this inference is a presupposition like the one in (50), projecting past negation.

Furthermore, to the extent we can have unrealizable desires, there is a contrast between the NOM and the ACC DP in the following example, with the ACC DP giving rise to a presupposition failure since its referent is unable of causing the eventuality of the embedded clause.¹⁵

- (52) Thelo o Yanis /#ton Yani na stamatisi ton chrono, ala
 want the.NOM Yanis.NOM the.ACC Yanis.ACC SBJV stop the time but
 ksero oti afto ine adhinato!
 know COMP this is impossible

‘I want Yanis to stop the time, but I know this is impossible!’

The infelicity persists under negation, suggesting that the presupposition projects:

¹⁵I thank an anonymous reviewer for urging me to think of such examples.

- (53) Dhen thelo o Yanis /#ton Yani na stamatisi ton chrono,
 NEG want the.NOM Yanis.NOM the.ACC Yanis.ACC SBJV stop the time
 jiati ksero oti afto ine adhinato!
 because know COMP this is impossible
 ‘I don’t want Yanis to stop the time, because I know this is impossible!’

Finally, notice that the presupposition is not challengeable, indicating that it is not at issue:

- (54) *Yanis is not a great student, but I really want him to try and study hard in order to succeed in the centralized university entrance exam.*

- a. A: Thelo ton Yanis na petichi pro stis eksetasis.
 want.PRS.1SG the.ACC Yanis.ACC SBJV succeed.3SG 3SG in-the exams

‘I want Yanis to succeed in the exams.’

- b. B: #Ma o Yianis dhen bori na to prokalesi afto!
 but the.NOM Yianis.NOM NEG can SBJV CL cause this

Intended: ‘But Yianis cannot cause this!’

- c. C: Ma dhen prokite na dhiavasi, ton kseris!
 but NEG going-to SBJV study him.ACC know.2SG

‘But he’s not going to study, you know him!’

Having motivated the presupposition of the proposed concealed causative, let me now turn to the assertion. What CAUSE asserts is that there is an event of the type described by the embedded clause such that the referent of the ACC DP is the causer of the event. A sentence with the concealed causative can thus be made false if the the NP-complement of the ACC DP does not in fact cause the eventuality described by the embedded clause. Consider the following scenario:

- (55) *I am the head of a political organization, and I want to encourage one of our members, Yanis, to be more courageous and fight with the police. So, before a demonstration I say:*

Thelo ton Yani [na silifthi pro].
 want.PRS.1SG the.ACC Yani.ACC SBJV arrest.PASS.3SG 3SG

‘I want Yanis to cause himself to get arrested.’

The concealed causative analysis predicts that my desire when uttering the sentence above is that Yanis causes himself to get arrested. As a reviewer points out, this further predicts that my desire is not satisfied in a scenario where Yanis does nothing to cause his arrest but is accidentally arrested by pure luck. Indeed, in such a scenario my desire is not satisfied,

since Yanis did not cause his arrest. This can be minimally contrasted with the NOM DP, where my desire is satisfied in any scenario where Yanis ends up arrested (either via doing something or by pure luck):

- (56) Thelo [o Yanis na silifthi].
 want.PRS.1SG the.NOM Yani.NOM SBJV arrest.PASS.3SG
 ‘I want Yanis to get arrested.’

Similarly, when uttering (54a) above, my desire would not be satisfied in a scenario where Yanis does not study or try at all, but he succeeds in the exam by accident (e.g., because the secretary who was writing down the results made a mistake giving him a higher score than his actual score).

As noted in the beginning of this Section, the fact that the construction with the ACC DP has different semantics than the one with the NOM DP was previously noted in the literature. Kotzoglou (2002) presents the following example:

- (57) Thelo ti mana /i mana su na min anakatevete
 want.1SG the.ACC mom.ACC the.NOM mom.NOM yours SBJV NEG interfere.PASS.3SG
 sta prosopika mas.
 in-the personal-affairs ours
 ‘I want your mom to not interfere in our personal affairs.’ Kotzoglou, 2002: ex.3(a), 3(b), p.41

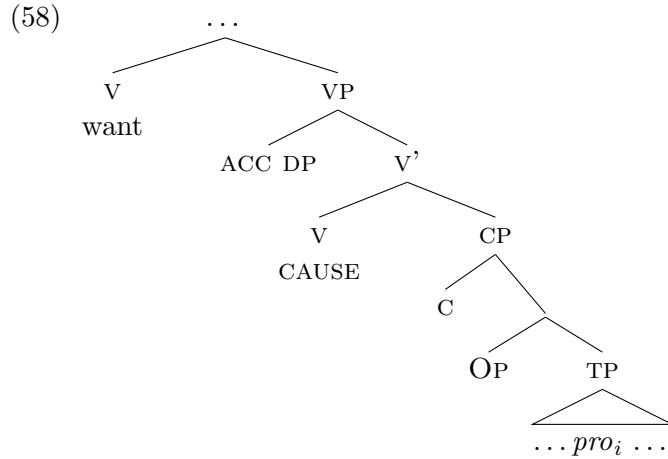
As Kotzoglou, 2002: p.41 notes in the version with the ACC DP “the speaker does not merely demand something to happen, but he actually demands that the person denoted by the DP* does or does not do the action described by the embedded verb” (where DP* stands for ACC DP). Indeed, in the example above, when the ACC DP is used, the desire would not be satisfied in a scenario where the mother does nothing to stop interfering but rather the couple simply moves to another country so she is unable to interfere. On the contrary, if the NOM DP is used the desire is satisfied in all scenarios where the mother (by doing something or by accident) stops interfering. In other words, the intuition described by Kotzoglou (2002) is in line with my argument that there is a concealed causative in the ACC DP version. All these examples suggest that when the ACC DP is used, there is a concealed causative CAUSE between the ACC DP and the embedded clause. In the following Section, I formalize this intuition presenting a hidden clause analysis of the construction.

6 Hidden clause analysis

Let me summarize what we have established so far about the construction. From a syntactic point of view, I have argued that the ACC DP is syntactically not part of the lowest clause. From a semantic point of view, I have argued that it is part of the intensionalized argument of the attitude verb, since it can have *de dicto* readings, and I also argued that it has causal

semantics. I propose to solve the intensionality paradox while also accounting for the causal semantics by arguing that there are three instead of two clauses. More specifically, I argue that there is a third, intermediate clause with a silent verb CAUSE that takes the ACC DP as its subject and the embedded clause as its complement. This will not only explain the causal semantics of the construction, but it will also solve the intensionality paradox introduced earlier on in the paper, since the NP-complement of the ACC DP will now be in the scope of the attitude verb.

I propose that there is a covert proleptic CAUSE verb, which takes the ACC DP as its subject and the embedded clause as its complement, as follows:¹⁶



This immediately solves the intensionality paradox, since *want* actually takes the VP clause headed by CAUSE as its argument, and therefore the ACC DP is part of the intensionalized argument of the verb; the *de dicto* readings are thus expected. In what follows I explain more in detail how this analysis works, as well as its theoretical motivations.

The hidden clause analysis proposed above is inspired by [den Dikken et al. \(2018\)](#), who analyze intensional transitive verbs (e.g., *wants* in ‘Johnny wants a unicorn’) not as transitives, despite surface appearances, but rather as clausal complement constructions. For example, they analyze the intensional transitives *imagine*, *want*, *need*, *look for* as taking a CP rather than a DP as their complement:

(59) Max imagined/wanted/needed/looked-for [*CP*... [*DP* a unicorn]...].

[den Dikken et al., 2018](#): based on p.47, ex.4(b)

More specifically, it is argued that *want*-type intensional transitive verbs contain a covert verb HAVE, part of a hidden clause:

¹⁶There might be an additional projection above the TP related to mood, but we abstract away from this for the purposes of this paper.

(60) I want TO HAVE a sister/ freedom/ a driveway.

den Dikken et al., 2018: based on p.48, ex.6

My proposal is similar in that I will also propose a hidden clause to solve the intensionality paradox and to capture the causal semantic requirement. The core idea behind both analyses is that what appears to be an intensionalized DP actually belongs to the clausal intensionalized argument of the attitude verb. In other words, both constructions suggest that there are hidden clausal layers solving intensionality paradoxes.

Going back to my proposal in (58), the verb *want* takes a small VP clause as its complement, containing the ACC DP as the subject of CAUSE, which in turn takes the embedded clause as its complement. Given that the VP clause is smaller than a TP or a CP, there is a sense in which we have two rather than three clauses, if we take a clause to be a full TP or a CP containing it. This phenomenon is akin to *restructuring* (Wurmbrand, 2002) in that the intermediate clause is small enough to lack a CP boundary between the matrix verb and the covert verb. Therefore, the DP gets its case from the matrix verb (possibly by further moving to the matrix clause, as suggested by a reviewer), which explains its accusative case. What is more, the covert verb cannot be pronounced either because it morphologically lacks tense and all pronounced verbs in MG are morphologically (but not necessarily syntactically, e.g., Terzi 1992; Iatridou 1993; Varlokosta 1994; Roussou 2009; Grano 2015) tensed, or because there's no Vocabulary Insertion rule for it.¹⁷ Finally, the operator that we see at the specifier of the CP is what makes CAUSE proleptic, guaranteeing that the ACC DP will obligatorily co-refer with a pronoun in the most embedded clause.

This operator is taken from Salzmann's (2017a) analysis of German prolepsis. In prolepsis, as described in Section 3, the proleptic object is obligatorily semantically related to a pronoun inside the embedded clause. Indeed, in the MG construction as well, the ACC DP needs to be semantically related to a pronoun in an argument position in the embedded clause. In the absence of such a pronoun the sentence becomes ungrammatical:

(61) *Thelo ton Dimitri [na kimithi to moro stis 9].
want.1SG the.ACC Dimitris.ACC SBJV sleep.PASS.3SG DET baby at 9

Intended: 'What I want of Dimitris is that the baby sleeps at 9.'

The ungrammaticality of the sentence above persists even if Dimitris is the babysitter of the baby, in which case it would be perfectly normal to expect him to cause the baby to sleep.¹⁸

¹⁷In a language with infinitives, a similar covert causative could be pronounceable.

¹⁸This meaning would instead be expressed via a PP containing the ACC DP.

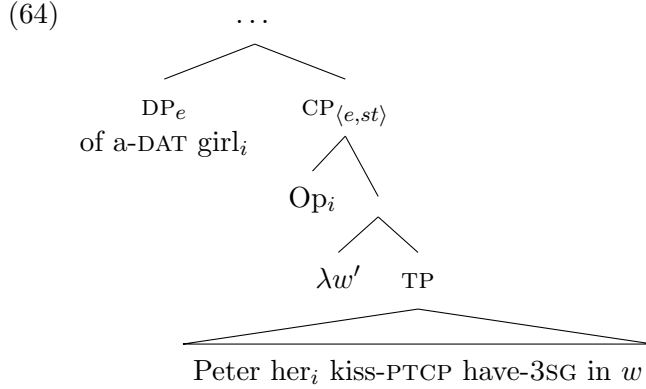
(62) Thelo apo ton Dimitri [na kimithi to moro stis 9].
want.1SG from the.ACC Dimitris.ACC SBJV sleep.PASS.3SG DET baby at 9
'I want from Dimitris (to make sure) that the baby sleeps at 9.'

To prevent CAUSE from being inserted above a CP with no (semantically related with the small VP clause's subject) pronoun, I argue that CAUSE is proleptic. [Salzmann \(2017a\)](#), in his analysis of German prolepsis, forces the proleptic object to co-refer with a pronoun in the embedded CP by making the CP predicative (of type $\langle e, st \rangle$). This means that the CP in (32), repeated below, semantically needs the proleptic object *von einem mädchen* as an argument.

- (63) **Von einem Mädchen** weiss ich, dass Peter es geküsst hat.
 Of a-DAT girl know-1SG-PRES I COMP Peter her kiss-PTCP have-3SG
 'Of a girl, I know that Peter kissed her.'

[Salzmann, 2017a](#): p.15, ex.30(a)

The CP becomes predicative thanks to an operator in SpecCP which abstracts over a pronoun, making the CP of type $\langle e, st \rangle$, as follows:

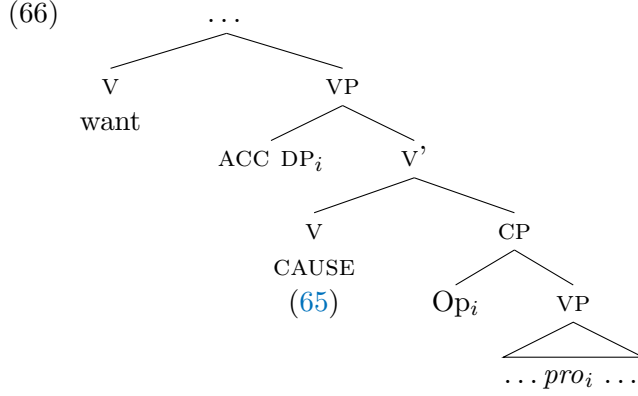


I adopt the same solution to force the obligatory co-reference between the ACC DP and a pronoun in the most embedded CP in the MG construction. I thus propose the following proleptic CAUSE, repeated from (50), which takes a *predicative* CP of type $\langle e, \langle v, \langle s, t \rangle \rangle \rangle$ as an argument.

- (65) $\llbracket \text{CAUSE} \rrbracket = \lambda P_{\langle e, \langle v, \langle s, t \rangle \rangle \rangle} . \lambda x_e . \lambda w_s . \exists e' \exists w' : P(x)(e')(w') \ \& \ \text{cause}(x, e'). \ \exists e [P(x)(e)(w) \ \& \ \text{cause}(x, e)]$

Thus, an intermediate small VP clause with a silent proleptic CAUSE verb solves the intentionality paradox, while also accounting for the semantic profile of the construction. Let me illustrate how this works.

More precisely, here is how we get the *de dicto* reading, without positing any extra denotation for *want* or assuming a semantic lowering mechanism, contra [Tsilia \(2023\)](#). We assume the following LF based on the proposed syntax in (58):



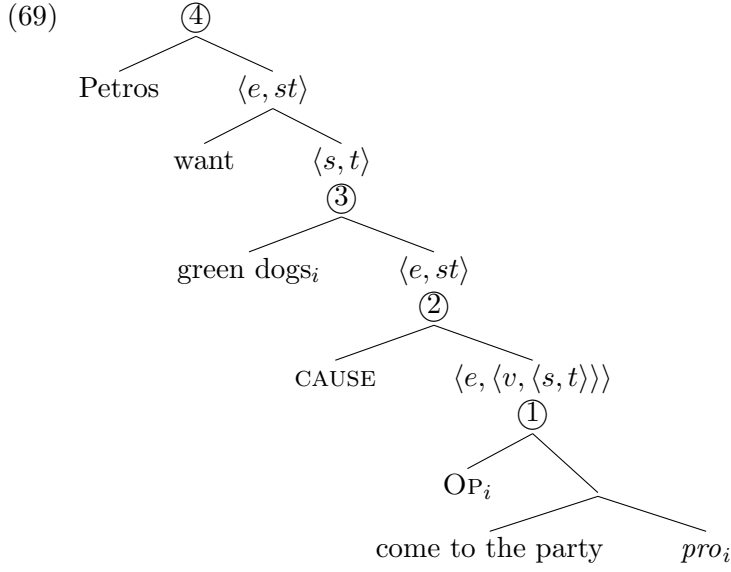
We simply use the regular entry for *want*, namely that the proposition it takes is true in all the worlds compatible with the subject's bouletic alternatives.¹⁹

$$(67) \llbracket \text{want} \rrbracket = \lambda p_{\langle s, t \rangle}. \lambda x. \lambda w. \forall w' \in \text{BOUL}(x, w) : p(w') = 1$$

The derivation of the *de dicto* reading of (41a), repeated as (68), proceeds as in LF (69), with CAUSE indicating that its subject causes the eventuality described by the embedded clause.

- (68) O Petrakis theli **prasinus skilus_i** [na erthun *pro_i*
 The.NOM Petros.DIM.NOM want.PRS.3SG green.ACC dogs.ACC SBJV come.3PL 3PL
 sto parti tu].
 to-the party his.GEN
 ‘Little Petros wants green dogs to come to his party.’

¹⁹The entry provided could be seen as an over-simplification. Whether *want* actually involves universal quantification over bouletic alternatives is not obvious (see a.o. Heim (1992); von Stechow (1999)). However, a precise lexical entry for *want* is not a desideratum of this paper, and this choice does not affect the analysis. I refer the interested reader among others to Heim (1992); von Stechow (1999); Pasternak (2019); Portner and Rubinstein (2020); Staniszewski (2022) and references therein.



The entry for CAUSE is in (65), and for *want* in (67). As for the predicate *come to the party*, it takes an individual, an event, and a world, and denotes that there is an event e where x comes to the party in w :

$$(70) \llbracket \text{come to the party} \rrbracket = \lambda x. \lambda e. \lambda w. \text{come-party}(x)(e)(w)$$

I treat *green dogs* as an existential quantifier, with the following lexical entry:

$$(71) \llbracket \text{green dogs} \rrbracket = \lambda P_{\langle e, \langle s, t \rangle \rangle}. \lambda w. \exists x [\text{green-dogs}(x)(w) \ \& \ P(x)(w)]$$

The operator, OP, abstracts over the pronoun in the embedded clause, resulting in the following denotation for node ①:

$$(72) \llbracket \text{①} \rrbracket = \lambda x. \lambda e. \lambda w. \text{come-party}(x)(e)(w)$$

Then this is fed into CAUSE, which existentially quantifies over the event, resulting in the following:

$$(73) \llbracket \text{②} \rrbracket = \lambda x. \lambda w. \exists e' \exists w' : \text{come-party}(x)(e')(w') \ \& \ \text{cause}(x, e'). \ \exists e [\text{come}(x)(e)(w) \ \& \ \text{cause}(x, e)]$$

Then, this composes with *green dogs*, and we end up with the following meaning:

$$(74) \llbracket \text{③} \rrbracket = \lambda w. \exists x [\text{green-dogs}(x)(w) \ \& \ \exists e' \exists w' : \text{come-party}(x)(e')(w') \ \& \ \text{cause}(x, e'). \ \exists e [\text{come}(x)(e)(w) \ \& \ \text{cause}(x, e)]]$$

After this composes with the attitude verb, we have:

$$(75) \llbracket \textcircled{4} \rrbracket = \lambda w. \forall w'' \in \text{BOUL}(\text{Petros}, w). \exists x [\text{green-dogs}(x)(w'') \ \& \ \exists e' \exists w' : \text{come-party}(x)(e')(w') \ \& \ \text{cause}(x, e'). \exists e [\text{come}(x)(e)(w'') \ \& \ \text{cause}(x, e)]]$$

If we evaluate this at the actual world (notated as w_0), we end up with the desired truth conditions for (68), namely that it is true if and only if in every world compatible with little Petros' bouletic alternatives green dogs cause the event that they come to the party:

$$(76) \forall w'' \in \text{BOUL}(\text{Petros}, w_0). \exists x [\text{green-dogs}(x)(w'') \ \& \ \exists e' \exists w' : \text{come-party}(x)(e')(w') \ \& \ \text{cause}(x, e'). \exists e [\text{come}(x)(e)(w'') \ \& \ \text{cause}(x, e)]]$$

We also have a presupposition that this is an event green dogs are able of causing. This sentence ends up meaning that what little Petros wants is for the dogs to *cause* their coming to the party. Thus, we see how the intermediate VP clause gets us the *de dicto* readings for free thanks to the referent of the ACC DP being part of the proposition the attitude verb takes as its argument.

As for the *de re* reading of (42), repeated as (77), it is derived by movement, with the following LF:²⁰

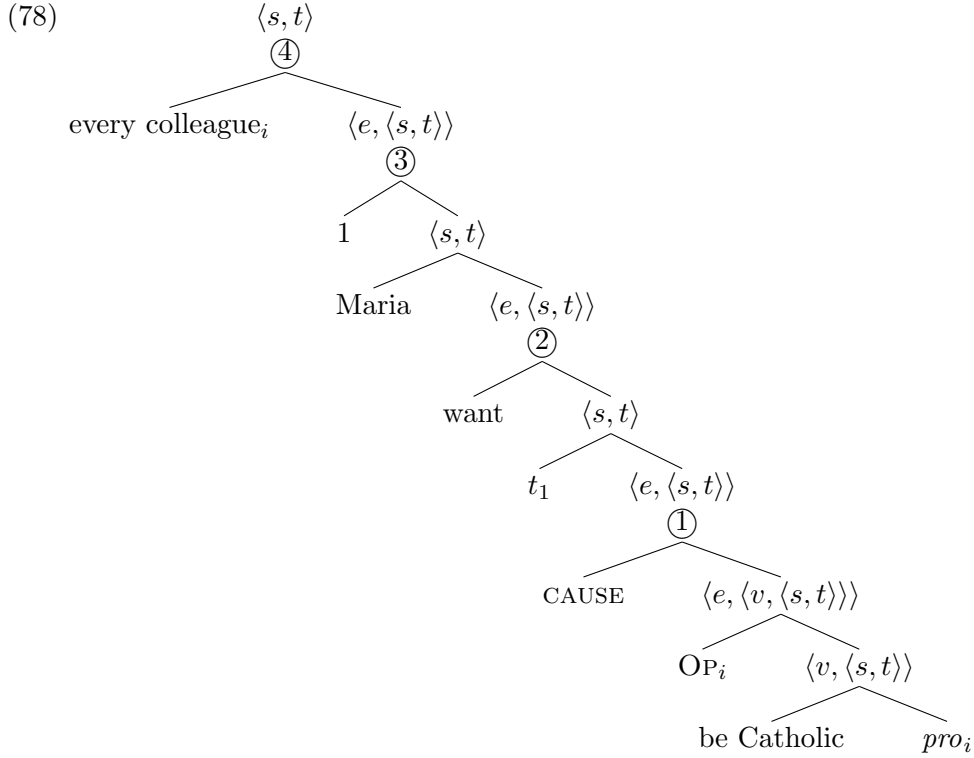
(77) *Maria is on an apostolic mission in Egypt during the pandemic, while working remotely. She started this job during COVID and thus never got to meet her colleagues, Markos and Christos. It just so happens that Markos and Christos are also in Egypt and Maria has met them without knowing they are her colleagues. She tried to convince them to become Catholic.*

I Maria theli **kathe** **tis** **sinadhelfo**_i [na ine *pro*_i
The.NOM Maria want.PRS every.ACC she.GEN colleague.ACC SBJV be.3SG 3SG
katholikos].
catholic.NOM

‘Maria wants every colleague of hers to be Catholic.’

Tsilia, 2023: p.665, ex.31

²⁰In the following tree, I QR the quantificational DP *every colleague*, and the denotation of the trace its given by the assignment function g .



We have the following new lexical entries:

$$(79) \llbracket \text{be Catholic} \rrbracket = \lambda x. \lambda e. \lambda w. \text{be-Catholic}(x)(e)(w)$$

$$(80) \llbracket \text{every colleague} \rrbracket = \lambda P_{\langle e, \langle s, t \rangle \rangle}. \lambda w. \forall x. \text{colleague}(x)(w) \rightarrow P(x)(w)$$

Now, let's calculate the denotation for the circled nodes:

- (81) a. $\llbracket \textcircled{1} \rrbracket = \lambda x. \lambda w. \exists e' \exists w' : \text{be-Catholic}(x)(e')(w') \ \& \ \text{cause}(x, e'). \exists e [\text{be-Catholic}(x)(e)(w) \ \& \ \text{cause}(x, e)]$
- b. $\llbracket \textcircled{2} \rrbracket = \lambda x. \lambda w. \forall w'' \in \text{BOUL}(x, w). \exists e' \exists w' : \text{be-Catholic}(g(1))(e')(w') \ \& \ \text{cause}(g(1), e'). \exists e [\text{be-Catholic}(g(1))(e)(w') \ \& \ \text{cause}(g(1), e)]$
- c. $\llbracket \textcircled{3} \rrbracket = \lambda x. \lambda w. \forall w'' \in \text{BOUL}(\text{Maria}, w). \exists e' \exists w' : \text{be-Catholic}(x)(e')(w') \ \& \ \text{cause}(x, e'). \exists e [\text{be-Catholic}(x)(e)(w') \ \& \ \text{cause}(x, e)]$
- d. $\llbracket \textcircled{4} \rrbracket = \lambda w. \forall x. \text{colleague}(x)(w) \rightarrow \forall w'' \in \text{BOUL}(\text{Maria}, w). \exists e' \exists w' : \text{be-Catholic}(x)(e')(w') \ \& \ \text{cause}(x, e'). \exists e [\text{be-Catholic}(x)(e)(w'') \ \& \ \text{cause}(x, e)]$

We thus end up with the desired meaning that for all colleagues in the actual world, Maria's desire is that they are Catholic (in her desire worlds) and that they caused themselves to

be so:

- (82) $\forall x. \text{colleague}(x)(w_0) \rightarrow \forall w'' \in \text{BOUL}(\text{Maria}, w_0).$
 $\exists e' \exists w' : \text{be-Catholic}(x)(e')(w') \ \& \ \text{cause}(x, e'). \ \exists e [\text{be-Catholic}(x)(e)(w'') \ \& \ \text{cause}(x, e)]$

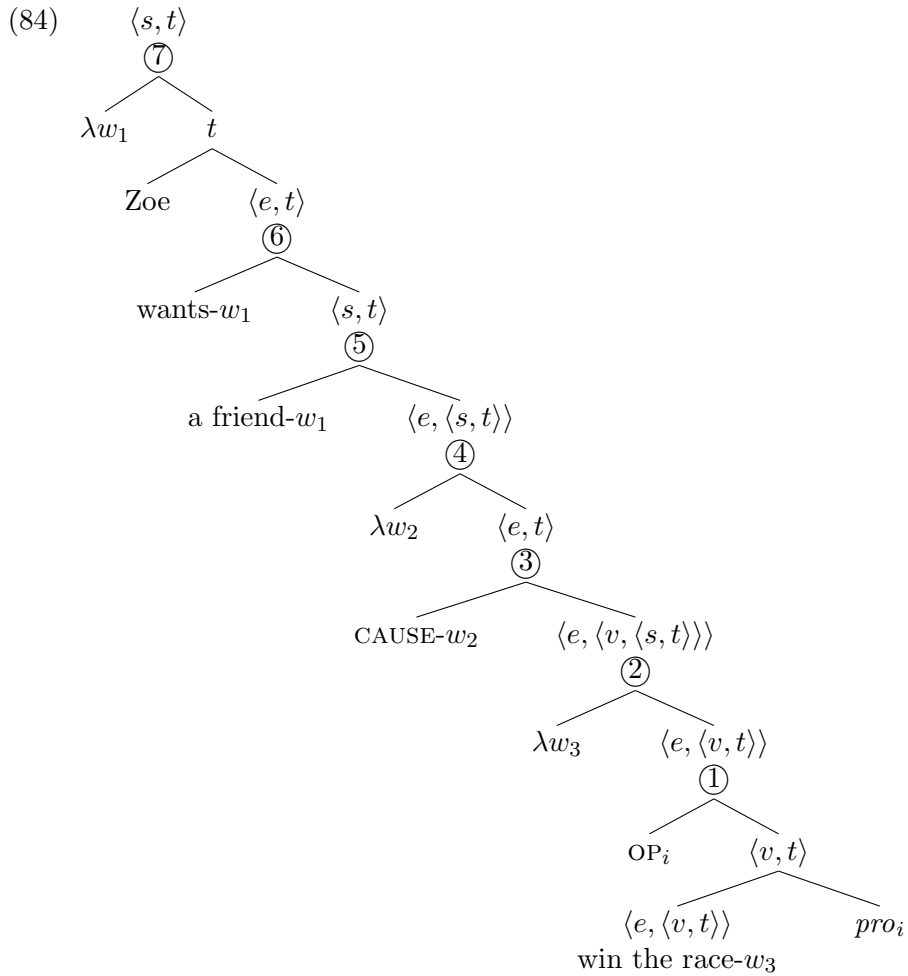
Finally, the third reading of (43), repeated as (83), can be derived as well, following the standard solution to the third reading by Percus (2000). That is, we assume the existence of world variables in the syntax; these world variables come with an index. All predicates have a world variable, and the index of their world variable need not match the index of the main predicate. Lambda abstractors bind the indexed world variables, as shown in the LF in (84).

- (83) *Zoi is attending a 100m race at the Olympics. Three contestants are talking to each other before the start. Unbeknownst to Zoi, these three contestants are my friends. She thinks to herself that she wants one of those three people to win the race, because they seem motivated.*

I Zoi theli [enan filo mu]_i [na kerdhisi pro_i ton
The.NOM Zoi want.PRS.3SG a.ACC friend.ACC mine SBJV win.3SG 3SG the.ACC
aghona].
race.ACC

‘Zoi wants a friend of mine to win the race.’

[Tsilia, 2023](#): p.665, ex.32, *pro* mine



The major constituents in the embedded clause are interpreted as follows:

(85) a. $\llbracket \text{win the race-}w_3 \rrbracket = \lambda x.\lambda e.\text{win-race}(x)(e)(w_3)$
b. $\llbracket \text{a friend-}w_1 \rrbracket = \lambda P_{\langle e, \langle s, t \rangle \rangle}.\lambda w.\exists x.\text{friend}(x)(w_1) \ \& \ P(x)(w)$

Since, CAUSE, and *want* now have world arguments, we also have:

(86) a. $\llbracket \text{CAUSE-}w_2 \rrbracket = \lambda P_{\langle e, \langle v, \langle s, t \rangle \rangle \rangle}. \lambda x. \exists e' \exists w' : P(x)(e')(w') \ \& \ \text{cause}(x, e'). \ \exists e [P(x)(e)(w_2) \ \& \ \text{cause}(x, e)]$

b. $\llbracket \text{want-}w_1 \rrbracket = \lambda p_{\langle s, t \rangle}. \lambda x. \forall w'' \in \text{BOUL}(x, w_1). \ p(w'')$

With that, let me give the denotations of the circled nodes:

- (87) a. $\llbracket \textcircled{1} \rrbracket = \lambda x. \lambda e. \text{win-race}(x)(e)(w_3)$
 b. $\llbracket \textcircled{2} \rrbracket = \lambda x. \lambda e. \lambda w_3. \text{win-race}(x)(e)(w_3)$
 c. $\llbracket \textcircled{3} \rrbracket = \lambda x. \exists e' \exists w' : \text{win-race}(x)(e')(w') \ \& \ \text{cause}(x, e').$
 $\exists e [\text{win-race}(x)(e)(w_2) \ \& \ \text{cause}(x, e)]$
 d. $\llbracket \textcircled{4} \rrbracket = \lambda x. \lambda w_2. \exists e' \exists w' : \text{win-race}(x)(e')(w') \ \& \ \text{cause}(x, e').$
 $\exists e [\text{win-race}(x)(e)(w_2) \ \& \ \text{cause}(x, e)]$
 e. $\llbracket \textcircled{5} \rrbracket = \lambda w. \exists x. \text{friend}(x)(w_1) \ \& \ \exists e' \exists w' : \text{win-race}(x)(e')(w') \ \& \ \text{cause}(x, e').$
 $\exists e [\text{win-race}(x)(e)(w) \ \& \ \text{cause}(x, e)]$
 f. $\llbracket \textcircled{6} \rrbracket = \lambda x. \forall w'' \in \text{BOUL}(x, w_1). \exists x. \text{friend}(x)(w_1) \ \& \ \exists e' \exists w' : \text{win-race}(x)(e')(w') \ \& \ \text{cause}(x, e').$
 $\exists e [\text{win-race}(x)(e)(w'') \ \& \ \text{cause}(x, e)]$
 g. $\llbracket \textcircled{7} \rrbracket = \lambda w_1. \forall w'' \in \text{BOUL}(\text{Zoe}, w_1). \exists x. \text{friend}(x)(w_1) \ \& \ \exists e' \exists w' : \text{win-race}(x)(e')(w') \ \& \ \text{cause}(x, e').$
 $\exists e [\text{win-race}(x)(e)(w'') \ \& \ \text{cause}(x, e)]$

We can thus also derive the third reading, since we get the desired truth conditions, namely that there is a friend in the actual world, such that in Zoe's bouletic alternatives they win the race and they cause themselves to do so. There is also the presupposition that they are able to cause such an event:

- (88) $\forall w'' \in \text{BOUL}(\text{Zoe}, w_0). \exists x. \text{friend}(x)(w_0) \ \& \ \exists e' \exists w' : \text{win-race}(x)(e')(w') \ \& \ \text{cause}(x, e').$
 $\exists e [\text{win-race}(x)(e)(w'') \ \& \ \text{cause}(x, e)]$

As a note, I assume that the class of verbs supporting this construction is identical to those selecting for CAUSE-VP. To my knowledge, there is no independent principle that would help us identify which verbs select for a CAUSE-VP, and thus, I am claiming that the set of verbs that participate in this construction is lexically idiosyncratic. This could also be the reason why some speakers find these constructions ungrammatical.

I thus proposed a solution to both the intensionality paradox and the causal semantic requirement with an intermediate small VP clause headed by proleptic CAUSE which in turn takes a predicative CP as an argument. This intermediate clause solves the intensionality paradox, accounting for opaque readings. The fact that CAUSE takes a predicative CP as its first argument explains the obligatory pronoun in the embedded clause, which has to be semantically related to the ACC DP, like in prolepsis. The question of what constrains the position of the pronoun inside the CP is left open for future research.

7 Evidence in favor of a hidden clause

Having presented the account, let me now turn to independent evidence for the existence of CAUSE. I will show that CAUSE can be modified by a temporal adverb. Crucially, *want* is atelic:

- (89) I Maria theli to amaksi tora jia/*se 30 lepta.
 the.NOM Maria.NOM want.PRS.3SG the.ACC car.ACC now for in 30 minutes.
 ‘Maria wants the car now for/*in 30 minutes.’

In the above example with *tora* ‘now’, only a *for*-temporal phrase is possible. Since *want* is atelic and cannot be modified by an *in*-temporal phrase, if we also pick an atelic verb in the CP, then the only telic verb to be modified by an *in*-temporal phrase would be CAUSE. I will show that we can modify CAUSE when the embedded verb is atelic, like *fonazo* ‘shout’:

- (90) O Dimitris fonazi to sinthima jia/*se mia ora.
 the.NOM Dimitris.NOM shout.PRS.3SG the.ACC slogan.ACC for in one hour.
 ‘Dimitris is shouting the slogan for/*in an hour.’

Having both a matrix and an embedded verb being atelic, the only way for an *in*-temporal phrase to modify something felicitously is if there is some hidden telic verb like CAUSE. Indeed, an *in*-temporal phrase is felicitous in the following:

- (91) *I am the leader of an anti-government political organization and Dimitris is a member. We are preparing for a protest, and I am saying that in five minutes when we start protesting I want Dimitris to start shouting the slogan for the next hour.*
 Lipon, thelo tora **ton** **Dimitri** /*o Dimitris se pente
 So, want.PRS.1SG now the.ACC Dimitri.ACC the.NOM Dimitri.NOM in five
 na fonazi *pro* to sinthima jia mia ora.
 SBJV shout.3SG 3SG the slogan for one hour
 ‘So, I now want Dimitris to cause it to be the case in 5 minutes that he shouts the slogan for an hour.’

The felicitousness of the above *in*-temporal phrase with the ACC DP reveals the existence of a telic silent verb, which I argue is CAUSE. When the DP is in the nominative and thus CAUSE is predicted to be absent, the sentence becomes ungrammatical.

Further evidence in favor of a silent CAUSE comes from PPs understood as modifying it. The following PP can logically only be modifying CAUSE:

- (92) *Yanis is a political activist and part of an organization run by me. I want to raise awareness about it and I think that getting someone arrested will give us some publicity to this end.*

Thelo [VP_Tton Yani_i me provokatsia [na ton_i silavi
 Want.PRS.1SG the.ACC Yanis.ACC with provocation SBJV him.ACC arrest.3SG
 i astinomia]].
 the.NOM police.NOM

‘I want Yanis via provocation to cause the police to arrest him.’

I thus presented independent evidence showing that we can modify a silent telic verb, i.e., CAUSE. As a control, I should note that if we have a NOM DP in (92) the sentence becomes ungrammatical, since there is no covert causal verb for the PP to modify:

- (93) *Yanis is a political activist and part of an organization run by me. I want to raise awareness about it and I think that getting someone arrested will give us some publicity to this end.*

*Thelo [o Yanis me provokatsia na ton silavi
 Want.PRS.1SG the.NOM Yanis.NOM with provocation SBJV him.ACC arrest.3SG
 i astinomia].
 the.NOM police.NOM

‘I want Yanis via provocation to cause the police to arrest him.’

8 Concealed causatives and direct causation

In the semantic literature on concealed causatives, there is a much-discussed observation that concealed and lexical causatives can only express direct causation. Let us separate causative constructions into (i) periphrastic causatives containing an overtly pronounced causative predicate as in (94a), (ii) lexical causatives like *kill* (i.e., *cause to die*) containing a *cause* in their denotation as in (94b), and (iii) concealed causatives which do not so obviously contain a *cause* as in (94c).

- (94) a. **Periphrastic causative:** John caused [the robber to die] [by shooting her].
 b. **Lexical causative:** John killed the robber [by shooting her].
 c. **Concealed causative:** John [shot] [the robber dead].

(Bittner, 1998: p.1, ex.1, 2)

Originally noted for periphrastic vs. lexical causatives (Fodor, 1970b), and then extended to concealed causatives as well (Dowty, 1979), the observation is that only periphrastic causatives can describe a situation of indirect causation. To see this, let me use Bittner’s scenario below:

- (95) There is a holdup in a bank. John shoots the robber, but he is not much of a shot. His bullet only grazes the robber’s ear, a trivial wound which unfortunately brings on a heart attack that causes instant death.

(Bittner, 1998: adapted from scenario in p.4)

Although (94a) would perfectly describe the situation above, (94b) would be misleading, and (94c) clearly false. Bittner (1998) describes this in the following way:

- (96) **Concealed Causative Semantics:** If a causal relation is syntactically concealed (only its arguments are overtly expressed), then it is semantically direct (no intermediate causes).

(Bittner, 1998: C, p.4)

Similarly, there are scenarios where a periphrastic but not a lexical causative would be used:

- (97) Suppose that the sheriff of an old West town is to fight a gun-duel with an infamous badman at high noon. Suppose also that, so as not to take any unnecessary chances, the sheriff goes to the local gunsmith to have his trusty sixshooter put in top working condition. Suppose, furthermore, that the gunsmith, who is a friend of the outlaw, installs an old, rusty firing pin in the sheriff’s gun and tells him that the newest and best available pin has been put in. Now, when the gun-duel takes place, the sheriff, who draws first, is unable to fire his gun because the defective firing pin prevents it from discharging, and the outlaw then shoots and kills the sheriff. Clearly, the gunsmith caused the death of the sheriff, but, equally clearly, the gunsmith did not kill him.

(Katz, 1970: p.253, footnote 31)

Only the periphrastic causative accurately describes this:

- (98) a. ✓The gunsmith caused the sheriff to die.
b. ✗The gunsmith killed the sheriff.

We, thus, see that lexical causatives, at least in English, can only express direct causation (Fodor 1970b; Smith 1970; Katz 1970; Shibatani 1976; Dowty 1979; Levin and Hovav 1999; Wolff 2003; Hovav and Levin 2001 a.o.). Is the concealed causative in MG restricted to direct causation? The answer is ‘it depends’. Firstly, there is no universally accepted definition of ‘directness’ in the literature. Defining ‘directness’ for the purposes of this constraint is a debated matter. For example, it is unclear whether by ‘direct’ we mean that the caused event needs to be temporally contiguous to its cause (e.g., Fodor 1970b) or lack intermediate causes (Wolff, 2003). For an overview of definitions of ‘directness’ see Wolff, 2003: 1.1, p.3 and Martin (2018). Secondly, there are also authors arguing against such a constraint altogether, or showing how context-specific the judgments are (Danlos, 2001; Neeleman and

van de Koot, 2012; Martin, 2018; Ausensi, 2020). Arguing for or against this constraint, or debating the definition of ‘directness’ is beyond the scope of this paper.

However, to situate the MG construction in this debate, I would like to point out that there can be intermediate causes. We have already seen this with the arrest examples, where Yanis is provoking the police, which in turn causes his arrest (92). Whether this contradicts the generalization or not depends on our definition of ‘directness’. In a framework like Wolff’s 2003, provoking the police is not an intermediate cause, since it is an *enabling* condition (i.e., it *enables* the police to arrest him).²¹ What about an intermediate cause that is not an enabling condition? Consider the following:

- (99) Thelo ton Yanis [na petichi *pro* stis eksetasis].
 want the.ACC Yanis.ACC SBJV succeed.3SG 3SG in-the exams
 ‘I want Yanis to succeed in the exam.’

My desire above would be satisfied if he studies a lot and ends up succeeding (no intermediate cause scenario). But would my desire be satisfied if Yanis bribes the exam grader, does not study, does badly in the exam but ends up getting a good grade (intermediate cause scenario)? In that case, having uttered (99), my desire would not be satisfied. An overt *cause* would be more appropriate in the intermediate cause scenario:

- (100) Thelo [o Yanis na prokalesi tin epitichia tu stis
 want the.NOM Yanis.NOM SBJV cause.3SG the.ACC success.ACC his.GEN in-the
 eksetasis].
 exams
 ‘I want Yanis to cause his success in the exam.’

It thus seems that the concealed causative in question allows for intermediate causes only when they are enabling conditions in the sense of Wolff (2003). I leave a more thorough investigation of what the right notion of ‘directness’ is for the MG construction for future research.

9 Facing some issues

Lastly, I discuss some apparent issues with the account. First, I focus on a syntactic argument suggesting that the ACC DP is part of the matrix clause, showing that it does not contradict the proposed account, since the ACC DP receives its case from the matrix verb. Second, I discuss some *de dicto* limitations with strong quantifiers, also observed with intensional transitive verbs generally, posing a potential over-generation problem.

²¹For Wolff (2003) there should be no intermediate causes except if they are enabling conditions, where an intermediate entity *z* is an *enabling condition* if give a paraphrase of ‘*x* V-ed *y* by *z*’ as ‘*z* enabled *x* to V *y*’ (e.g., *x* killed *y* by *z* / *z* enabled *x* to kill *y*, see Martin, 2018: ft.10, p.111).

9.1 Syntactic behavior of PPs

Let me revisit an argument in favor of the ACC DP being part of the matrix clause. How can the ACC DP belong to the matrix clause (as argued in Section 2) if it is in reality part of an intermediate small VP clause? I will show that the two are compatible, since the ACC DP needs to get its case from the matrix verb.

In Section 2.1, we saw Kotzoglou’s (2002) observation that PPs can intervene between the ACC DP and the CP. There are two available parses for the PP: it can either modify the matrix verb as in (101a) or the embedded one as in (101b), where the PP occupies a topicalized or focused position in the left periphery. Crucially, the meaning changes; the habit ends up being the son’s (101a) or the father’s (101b) accordingly.

- (101) a. *Ithele ton patera tu apo sinithia [na tu lei*
 Want.PST.3SG the.ACC father.ACC he.GEN by habit SBJV he.GEN say.3SG
pro kathe mera ke ena dhiaforetiko paramithi].
 3SG every day and a.ACC different.ACC fairy-tale.ACC
 ‘By habit, he wanted his father to read to him a different fairy tale every day.’
 (son’s habit)
- b. *Ithele ton patera tu [apo sinithia na tu lei*
 Want.PST.3SG the.ACC father.ACC he.GEN by habit SBJV he.GEN say.3SG
pro kathe mera ke ena dhiaforetiko paramithi].
 3SG every day and a.ACC different.ACC fairy-tale.ACC
 ‘He wanted his father to read to him by habit a different fairy tale every day.’
 (father’s habit)

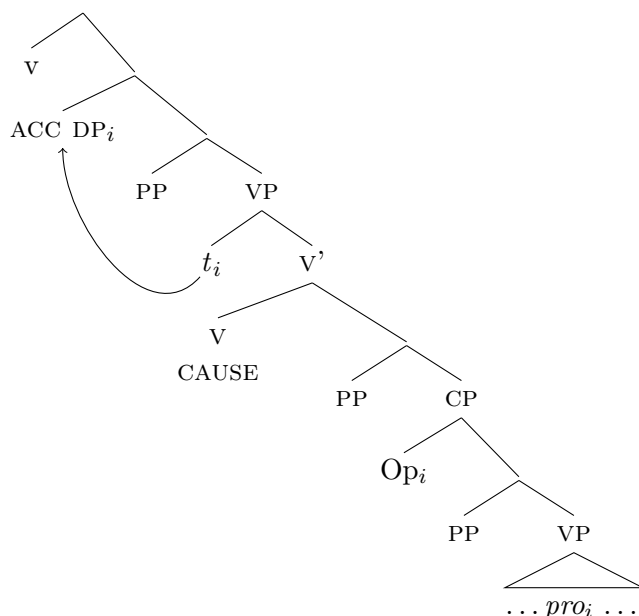
Kotzoglou, 2002: p.45, ex.7(b), brackets and *pro* are mine

Kotzoglou (2002) concludes that given that the ACC DP can precede a PP modifying the matrix verb as in (9a), it has to belong to the matrix clause – since the PP on its right is.

However, this does not contradict with the hidden clause analysis. The ACC DP could be born in an intermediate CAUSE clause and then move to the matrix clause. In other words, the fact that “by habit” in (101a) can modify the matrix verb does not necessarily entail that the ACC DP is not born in an intermediate clause. Movement of the ACC DP could be for case reasons, since as previously mentioned the ACC DP gets its case from the matrix verb. Alternatively, it could be an instance of scrambling, which we independently know MG has (see a.o. Drachman (1970); Philippaki-Warbuton (1985); Tzanidaki (1996); Alexiadou (1999); Alexiadou and Anagnostopoulou (2008); Kechagias (2008, 2011)). In other words, these data show that the ACC DP can move to the matrix clause (e.g., for case reasons) from its base-generated position as a subject of CAUSE. Thus, this does not pose a problem for the hidden clause analysis. In fact, as we saw in (92), a PP may be understood as modifying CAUSE, thus providing support for the hidden clause analysis. So PPs can attach to the

matrix verb, to the concealed causative, or to the embedded clause:

(102)



Movement of the ACC DP into the matrix clause would explain why a PP to the right of the ACC DP could be modifying the matrix verb, the causative, or the embedded clause.

9.2 De dicto limitations

I now turn to an over-generation problem. The proposed analysis straightforwardly accounts for *de dicto* readings by solving the intensionality paradox, since the ACC DP is part of the intensionalized argument of the attitude verb.

Could it be that the proposed analysis predicts too many *de dicto* readings? For intensional transitives too, Zimmermann (1993) argues that there are sometimes no *de dicto* readings with strong quantifiers:

(103) Alain is seeking each comic book.

Zimmermann, 1993: p.161, ex.27

(104) Alain is trying to find each comic book.

Zimmermann, 1993: p.161, ex.28

The first sentence has a *de dicto* reading in which Alain is searching with the general aim of finding every comic book, while the second only has the *de re* reading where for each comic book, Alain is searching for it. In the MG construction too, *de dicto* readings are degraded with strong quantifiers:

- (105) ??O Yanakis theli **kathe** prasino skilo [na erthi *pro*
 DET little-Yanis want.PRS.3SG every green dog.ACC SBJV come.3SG 3SG
 sto parti].
 to-the patry

‘Little Yanis wants every green dog to come to the party.’

A property analysis of intensional transitives predicts, following [Zimmermann \(1993\)](#), the absence of *de dicto* readings with strong quantifiers, since they need to be QRed.²² However, a clausal analysis of intensional transitives, following [den Dikken et al. \(2018\)](#), does not predict the infelicitousness of *de dicto* readings with strong quantifiers. Given that the current analysis is similarly a clausal analysis, it does not predict such infelicitousness either.

To the clausal analysis’ defense, it is not clear that we need to predict infelicitousness. There is disagreement in the literature with respect to whether intensional transitives have *de dicto* readings with strong quantifiers or not. [den Dikken et al. \(2018\)](#) argue that the *de dicto* reading is natural in the following:

- (106) a. Alain is seeking every spider in the house.

[den Dikken et al., 2018: p.77, ex.76\(a\)](#)

- b. Alain is seeking at most five spiders.

[den Dikken et al., 2018: p.77, ex.76\(b\)](#)

- c. I have looked for every typo in the manuscript.

[den Dikken et al., 2018: p.78, ex.77](#)

Putting this disagreement aside, and given that at least in (105) there is no *de dicto* reading with a strong quantifier, I need to account for the absence of such a reading. In the intensional transitives literature, [Moltmann \(1997\)](#) suggests an intensional quantifier analysis, which has the same over-generation problem as the clausal analysis. She rules out the unattested readings with an independent principle. This is meant to explain that, in line with [den Dikken et al. \(2018\)](#), some strong quantifiers can have *non*-specific readings (see (106)) but only if their NP-complement is read transparently, being evaluated with respect to the actual world. So, (105) is bad because ‘green dogs’ cannot be read transparently. Indeed, strong quantifiers with a transparently-read NP-complement are acceptable:

²²This is because in his analysis, strong quantifiers do not have the type of properties, as opposed to existential ones, which denote properties. I refer the reader to [Zimmermann \(1993\)](#) for details.

- (107) I diefthidria theli **kathe** mathiti tu scholiu [na ine *pro*
 DET principal want.PRS.3SG every student.ACC of school SBJV be.3SG 3SG
 charumenos].
 happy

‘The head principal wants every student in the school to be happy.’

Strong quantifiers thus require a domain of objects that exist in the actual world, although they do not have to be specific. To account for this [Moltmann \(1997\)](#) uses the following principle:

- (108) **Domain presupposition thesis:** Strong quantifiers presuppose their domain.

([Moltmann, 1997](#): p.25, ex.(56))

This rules out the unattested readings, and will also work for the MG construction, since it has the consequence that a strong quantifier “is acceptable in a context *c* only if the domain of the quantifier is included in the domain of *c* (the set of entities that are accepted in *c*)” ([Moltmann, 1997](#): p.25).

10 Conclusion

We explored the syntax and the semantics of an attitudinal construction in MG, where an attitude verb takes an ACC DP, semantically related to a pronoun in the subject or object position of the embedded clause. I showed that the ACC DP is base-generated above the lower clause syntactically but behaves as if it is part of the clause semantically, since it can have low-scope and opaque readings. I proposed a clausal analysis of the construction, according to which the ACC DP is in the scope of the attitude verb by virtue of being the subject of an intermediate small VP clause, headed by a silent proleptic CAUSE. I further motivated the choice of CAUSE by the causal semantics of the construction. Under the proposed analysis there is no mismatch at the syntax-semantics interface. The concealed causative intervening between the matrix and the embedded clause solves the intensionality paradox, while also accounting for the causal semantics.

In a nutshell, I explored a puzzling construction pointing to a mismatch at the syntax-semantics interface, and argued that it can be solved if we posit a concealed causative. This work contributes to the cross-linguistic investigation of proleptic constructions, concealed causatives, as well as to the cross-linguistic picture suggesting the existence of hidden clausal structure solving intensionality paradoxes, which in the case of MG is a hidden layer of c(l)ausality.

Statements and Declarations

The author has no conflict of interest to declare.

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